

Continuous-Time Fourier Series

Exclusive by smallTeamCreations

3.1 INTRODUCTION

As we have seen in the previous chapter, we can obtain the response of a linear system to an arbitrary input by representing it in terms of basic signals. The specific signals used were the shifted δ -functions. Often, it is convenient to choose a set of orthogonal waveforms as the basic signals. There are several reasons for doing this. First, it is mathematically convenient to represent an arbitrary signal as a weighted sum of orthogonal waveforms, since many of the calculations involving signals are simplified by using such a representation. Second, it is possible to visualize the signal as a vector in an orthogonal coordinate system, with the orthogonal waveforms being coordinates. Finally, representations in terms of orthogonal basis functions provides a convenient means of solving for the response of linear systems to arbitrary inputs.

For periodic signals, a convenient choice for an orthogonal basis is the set of harmonically related complex exponentials. The choice of these waveforms is appropriate since such complex exponentials are periodic, are relatively easy to manipulate mathematically, and yield results that have a meaningful physical interpretation. *The representation of a non-sinusoidal periodic signal in terms of complex exponentials, or equivalently, in terms of sine and cosine waveforms, leads to the Fourier series.* Or, in other words, *the Fourier series is a mathematical tool that allows the representation of any periodic signal as the sum of harmonically related sinusoids.* The Fourier series is named after the French physicist Joseph Fourier, who was the first to suggest that periodic signals could be represented by a sum of sinusoids.

So far, we have considered only time domain descriptions of continuous-time signals and systems. In this chapter, we introduce the concept of frequency domain representations and we learn how to decompose periodic signals into their frequency components.

3.1.1 A Vector View of Signals: Orthogonal Representations

Orthogonal representation of signals are important in solving many engineering problems. Two of the reasons this is so are that it is mathematically convenient to represent arbitrary signals as a weighted sum of orthogonal waveforms, since many of the calculations involving signals are simplified by using such a representation, and that it is possible to

visualize the signal as a vector in an orthogonal coordinate system, with the orthogonal waveforms being the unit coordinates.

We define an N -dimensional *orthogonal space* as a space characterized by a set of N linearly independent functions $\{\phi_k(t)\}$ called *basis functions*. Any arbitrary function in the space can be generated by a linear combinations of these functions. The basis functions must satisfy the conditions

$$\int_0^T \phi_k(t) \phi_m^*(t) dt = C_k \delta(k - m), \quad 0 \leq t \leq T \quad k, m = 1, 2, \dots, N \quad (3.1)$$

where $\phi_k^*(t)$ stands for the complex conjugate of the signal and $\delta(k - m)$, called the Kronecker delta function, is defined as

$$\delta(k - m) = \begin{cases} 1, & k = m \\ 0, & k \neq m \end{cases}$$

When the C_k constants are nonzero, the signal space is called orthogonal. When the basis functions are normalized so that each $C_k = 1$, the space is called an orthonormal space. The principal requirement for orthogonality can be stated as follows: Each $\phi_k(t)$ function of the set of basis functions must be independent of the other members of the set. From a geometric point of view, each $\phi_k(t)$ is mutually perpendicular to each of the other $\phi_k(t)$ for $k \neq m$. An example of such a space with $N = 3$ is shown in Fig. 3.1, where the mutually perpendicular axes are designated $\phi_1(t)$, $\phi_2(t)$, and $\phi_3(t)$. If $\phi_k(t)$ corresponds to a real-valued voltage or current waveform component, associated with a 1Ω resistive load, then the energy dissipated in the load in T seconds, due to $\phi_k(t)$, is

$$E_k = \int_0^T \phi_k^2(t) dt = C_k \quad (3.2)$$

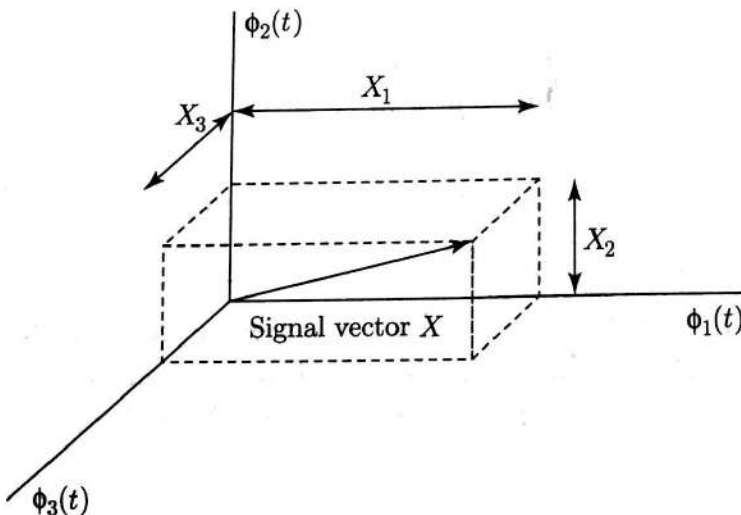


Fig. 3.1 Vectorial representation of the signal $x(t)$

Example 3.1 Show that the signals $\phi_m(t) = \sin mt$, $m = 1, 2, 3, \dots$, form an orthogonal set on the interval $-\pi < t < \pi$.

Solution

The signals $\phi_m(t) = \sin mt$, $i = 1, 2, 3, \dots$, form an orthogonal set on the interval $-\pi < t < \pi$ because

$$\begin{aligned} \int_{-\pi}^{\pi} \phi_m(t) \phi_n(t) dt &= \int_{-\pi}^{\pi} (\sin mt)(\sin nt) dt \\ &= \frac{1}{2} \int_{-\pi}^{\pi} \cos(m-n)t dt - \frac{1}{2} \int_{-\pi}^{\pi} \cos(m+n)t dt \\ &= \begin{cases} \pi, & m = n \\ 0, & m \neq n \end{cases} \end{aligned}$$

Since the energy in each signal equals π , the following set of signals constitutes an orthonormal set over the interval $-\pi < t < \pi$:

$$\frac{\sin t}{\sqrt{\pi}}, \frac{\sin 2t}{\sqrt{\pi}}, \frac{\sin 3t}{\sqrt{\pi}}, \dots$$

Example 3.2 Show that the signals $\phi_k(t) = e^{j(2\pi kt)/T}$, $k = 0, \pm 1, \pm 2, \dots$, form an orthogonal set on the interval $0 < t < T$.

Solution:

The signals $\phi_k(t) = e^{j(2\pi kt)/T}$, $k = 0, \pm 1, \pm 2, \dots$, form an orthogonal set on the interval $0 < t < T$ because

$$\begin{aligned} \int_{-\pi}^{\pi} \phi_k(t) \phi_m(t) dt &= \int_0^T e^{j(2\pi kt)/T} e^{-j(2\pi mt)/T} dt \\ &= \begin{cases} T, & k = m \\ 0, & k \neq m \end{cases} \end{aligned}$$

and hence, the signal $1/\sqrt{T} e^{j(2\pi kt)/T}$ constitutes an orthonormal set over the interval $0 < t < T$.

Orthonormal sets are useful in that they lead to a series representation of signals in a relatively simple fashion. Let $x(t)$ be a signal with finite energy over the interval $0 < t < T$. $x(t)$ can be expressed as a linear combination of N orthogonal waveforms $\phi_1(t), \phi_2(t), \dots, \phi_N(t)$ as

$$x(t) = X_1 \phi_1(t) + X_2 \phi_2(t) + \dots + X_N \phi_N(t)$$

This relationship is expressed in more compact notation as

$$x(t) = \sum_{k=1}^N X_k \phi_k(t) \quad (3.3)$$

where

$$X_k = \frac{1}{C_k} \int_0^T x(t) \phi_k(t) dt, \quad k = 1, \dots, N \quad (3.4)$$

The coefficient X_k is the value of the $\phi_k(t)$ component of signal $x(t)$. The signal $x(t)$ can be viewed as a set of vectors, $\mathbf{X} = \{X_1, X_2, \dots, X_N\}$. If, for example, $N = 3$, we may plot the vector $\mathbf{X}$ corresponding to the waveform

$$x(t) = X_1 \phi_1(t) + X_2 \phi_2(t) + X_3 \phi_3(t)$$

as a point in a three-dimensional Euclidean space with coordinates (X_1, X_2, X_3) as shown in Fig. 3.1. In general, once a set of N orthogonal functions has been adopted, $x(t)$ is completely determined by the vector of its coefficients:

$$\mathbf{X} = (X_1, X_2, \dots, X_N) \quad (3.5)$$

The normalized energy E_x of $x(t)$ over the interval $0 < t < T$ is

$$\begin{aligned} E_x &= \int_0^T |x(t)|^2 dt = \int_0^T \left| \sum_{k=1}^N X_k \phi_k(t) \right|^2 dt \\ &= \sum_{j=1}^N \sum_{k=1}^N X_j X_k^* \int_0^T \phi_j(t) \phi_k^*(t) dt \\ &= \sum_{j=1}^N \sum_{k=1}^N X_j X_k^* C_k \delta(j - k) \\ &= \sum_{k=1}^N |X_k|^2 C_k \end{aligned}$$

This result relates the energy of the signal $x(t)$ to the sum of the squares of the orthogonal series coefficients. If orthonormal functions are used (i.e., $C_k = 1$), the normalized energy is given by

$$E_x = \sum_{k=1}^N |X_k|^2$$

In terms of the coefficient vector $\mathbf{X}$, the normalized energy can be written as

$$E_x = (\mathbf{X}^*)^T \mathbf{X}$$

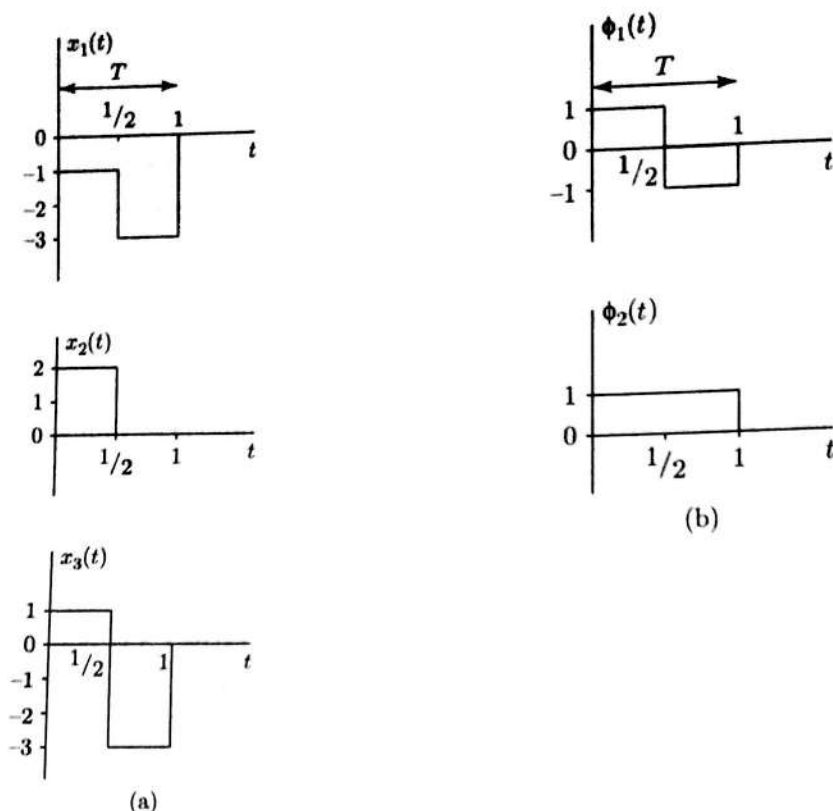


Fig. 3.2 (a) Arbitrary signal set and (b) a set of orthogonal basis functions

Example 3.3

- Figure 3.2(a) shows a set of three waveforms, $x_1(t)$, $x_2(t)$, and $x_3(t)$. Demonstrate that these waveforms do not form an orthogonal set.
- Figure 3.2(b) shows a set of two waveforms, $\phi_1(t)$ and $\phi_2(t)$. Verify that these waveforms form an orthogonal set.
- Show how the nonorthogonal waveforms set in part (a) can be expressed as a linear combination of the orthogonal set in part (b).

Solution

(a) $x_1(t)$, $x_2(t)$, and $x_3(t)$ are clearly not orthogonal, since they do not meet the requirements of Eq. (3.1); that is, the time integrated value (over a symbol duration) of the cross product of any two of the three waveforms is not zero. Let us verify this for $x_1(t)$ and $x_2(t)$:

$$\begin{aligned}
 \int_0^T x_1(t)x_2(t) dt &= \int_0^{T/2} x_1(t)x_2(t) dt + \int_{T/2}^T x_1(t)x_2(t) dt \\
 &= \int_0^{T/2} (-1)(2) dt + \int_{T/2}^T (-3)(0) dt \\
 &= -T = -1
 \end{aligned}$$

Similarly, for $x_2(t)$ and $x_3(t)$:

$$\begin{aligned}\int_0^T x_2(t)x_3(t) dt &= \int_0^{T/2} x_2(t)x_3(t) dt + \int_{T/2}^T x_2(t)x_3(t) dt \\ &= \int_0^{T/2} (2)(1) dt + \int_{T/2}^T (0)(-3) dt \\ &= T = 1\end{aligned}$$

Similarly, for $x_1(t)$ and $x_3(t)$:

$$\begin{aligned}\int_0^T x_1(t)x_3(t) dt &= \int_0^{T/2} x_1(t)x_3(t) dt + \int_{T/2}^T x_1(t)x_3(t) dt \\ &= \int_0^{T/2} (-1)(1) dt + \int_{T/2}^T (-3)(-3) dt = 4T = 4\end{aligned}$$

The integral over the interval T of each of the cross products $x_1(t)x_2(t)$, $x_2(t)x_3(t)$, and $x_1(t)x_3(t)$ results in nonzero values. Hence, the waveform set $\{x_i(t)\}(i = 1, 2, 3)$ in Fig. 3.2(a) is not an orthogonal set.

(b) Using Eq. (3.1), we verify that $\phi_1(t)$ and $\phi_2(t)$ form an orthogonal set as follows:

$$\int_0^T \phi_1(t)\phi_2(t) dt = \int_0^{T/2} (1)(1) dt + \int_{T/2}^T (-1)(1) dt = 0$$

The integral over the interval T of the cross product $\phi_1(t)\phi_2(t)$ results in zero value. Hence, the waveforms $\phi_1(t)\phi_2(t)$ in Fig. 3.2(b) form an orthogonal set.

(c) Using Eq. (3.2),

$$C_1 = \int_0^T \phi_1^2(t) dt = \int_0^{T/2} (1)^2 dt + \int_{T/2}^T (-1)^2 dt = T$$

and

$$C_2 = \int_0^T \phi_2^2(t) dt = \int_0^T (1)^2 dt = T$$

now using Eq. (3.4) with $C_1 = C_2 = T = 1$, we can express the nonorthogonal set $\{x_i(t)\}(i = 1, 2, 3)$ as a linear combination of the orthogonal basis waveforms $\phi_1(t)$

and $\phi_2(t)$:

$$x_1(t) = X_{11}\phi_1(t) + X_{12}\phi_2(t)$$

$$x_2(t) = X_{21}\phi_1(t) + X_{22}\phi_2(t)$$

$$x_3(t) = X_{31}\phi_1(t) + X_{32}\phi_2(t)$$

where, from Eq. (3.4), X_{1k} is

$$X_{1k} = \frac{1}{C_k} \int_0^T x_1(t)\phi_k(t) dt$$

Therefore,

$$X_{11} = \frac{1}{C_1} \int_0^T x_1(t)\phi_1(t) dt = \frac{1}{T} \int_0^{T/2} (-1)(1) dt + \int_{T/2}^T (-3)(-1) dt = 1$$

$$X_{12} = \frac{1}{C_2} \int_0^T x_1(t)\phi_2(t) dt = \frac{1}{T} \int_0^{T/2} (-1)(1) dt + \int_{T/2}^T (-3)(1) dt = -2$$

Now, from Eq. (3.4), X_{2k} is

$$X_{2k} = \frac{1}{C_k} \int_0^T x_2(t)\phi_k(t) dt$$

Therefore,

$$X_{21} = \frac{1}{C_1} \int_0^T x_2(t)\phi_1(t) dt = \frac{1}{T} \int_0^{T/2} (2)(1) dt + \int_{T/2}^T (0)(-1) dt = 1$$

$$X_{22} = \frac{1}{C_2} \int_0^T x_2(t)\phi_2(t) dt = \frac{1}{T} \int_0^{T/2} (2)(1) dt + \int_{T/2}^T (0)(1) dt = 1$$

Again, from Eq. (3.4), X_{3k} is

$$X_{3k} = \frac{1}{C_k} \int_0^T x_3(t)\phi_k(t) dt$$

Therefore,

$$X_{31} = \frac{1}{C_1} \int_0^T x_3(t) \phi_1(t) dt = \frac{1}{T} \int_0^{T/2} (1)(1) dt + \int_{T/2}^T (-3)(-1) dt = 2$$

$$X_{32} = \frac{1}{C_2} \int_0^T x_3(t) \phi_2(t) dt = \frac{1}{T} \int_0^{T/2} (1)(1) dt + \int_{T/2}^T (-3)(1) dt = -1$$

The nonorthogonal set $\{x_i(t)\}$ ($i = 1, 2, 3$) is a linear combination of the orthogonal basis waveforms $\phi_1(t)$ and $\phi_2(t)$ as

$$x_1(t) = \phi_1(t) - 2\phi_2(t)$$

$$x_2(t) = \phi_1(t) + \phi_2(t)$$

$$x_3(t) = 2\phi_1(t) - \phi_2(t)$$

3.2 FOURIER SERIES

The Fourier series is a mathematical tool that allows the representation of any periodic signal as the sum of harmonically related sinusoids. Any periodic signal, i.e., one for which $x(t) = x(t + T)$, can be expressed by a Fourier series provided that

1. if it is discontinuous, there are a finite number of discontinuities in the period T ;
2. it has a finite average value over the period T ;
3. it has a finite number of positive and negative maxima in the period T .

When these *Dirichlet conditions* are satisfied, the Fourier series exist. The Fourier series is of two types: (i) trigonometric Fourier series and (ii) exponential Fourier series.

3.2.1 Trigonometric Fourier Series

The trigonometric Fourier series is expressed as

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)] \quad (3.6)$$

where a_0 , a_n , and b_n are the trigonometric Fourier series coefficients. The coefficients may be obtained from $x(t)$ using

$$a_0 = \frac{1}{T} \int_0^T x(t) dt \quad (3.7)$$

$$a_n = \frac{2}{T} \int_0^T x(t) \cos(n\omega_0 t) dt \quad (3.8)$$

$$b_n = \frac{2}{T} \int_0^T x(t) \sin(n\omega_0 t) dt \quad (3.9)$$

The integrations can be carried out from $-T/2$ to $T/2$ or over any other full period that might simplify the calculation. The series with coefficients obtained from the above evaluation integrals converges uniformly to the function at all points of continuity and converges to the mean value at points of discontinuity. The series of Eq. (3.6) may be written in a number of apparently different although equivalent forms (polar form).

3.2.2 Polar Form Representation of the Fourier series

There are two ways to represent the Fourier series in a polar form:

Case I Assume $a_n = c_n \cos(\theta_n)$ and $b_n = -c_n \sin(\theta_n)$, where c_n and θ_n are related to a_n and b_n as

$$c_0 = a_0 \quad \text{and} \quad c_n = \sqrt{a_n^2 + b_n^2}, \quad \text{for } n \geq 1 \quad (3.10)$$

www.dahiyastech.blogspot.com

$$\theta_n = \tan^{-1} \frac{-b_n}{a_n} \quad (3.11)$$

Substituting $a_n = c_n \cos(\theta_n)$ and $b_n = -c_n \sin(\theta_n)$ into Eq. (3.6) gives

$$\begin{aligned} x(t) &= a_0 + \sum_{n=1}^{\infty} [c_n \cos(\theta_n) \cos(n\omega_0 t) - c_n \sin(\theta_n) \sin(n\omega_0 t)] \\ &= a_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t + \theta_n) \\ x(t) &= c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t + \theta_n) \end{aligned} \quad (3.12)$$

with c_n and θ_n defined by Eqs (3.10) and (3.11). The coefficient c_n is the amplitude and θ_n the phase of the n th harmonic. Observe that if we know that a Fourier series is to be constructed in the form of Eq. (3.12), then the set of numbers c_n and θ_n contains all the needed information. The plot of c_n as a function of n is known as the amplitude spectrum; the plot of θ_n as a function of n is the phase spectrum.

Case II Assume $a_n = c_n \sin(\phi_n)$ and $b_n = c_n \cos(\phi_n)$, where c_n and ϕ_n are related to a_n and b_n as

$$c_0 = a_0 \quad \text{and} \quad c_n = \sqrt{a_n^2 + b_n^2}, \quad \text{for } n \geq 1 \quad (3.13)$$

$$\phi_n = \tan^{-1} \frac{a_n}{b_n} \quad (3.14)$$

Substituting $a_n = c_n \sin(\phi_n)$ and $b_n = c_n \cos(\phi_n)$ in Eq. (3.6) gives

$$\begin{aligned}
 x(t) &= a_0 + \sum_{n=1}^{\infty} [c_n \sin(\phi_n) \cos(n\omega_0 t) + c_n \cos(\phi_n) \sin(n\omega_0 t)] \\
 &= a_0 + \sum_{n=1}^{\infty} c_n \sin(n\omega_0 t + \phi_n) \\
 x(t) &= c_0 + \sum_{n=1}^{\infty} c_n \sin(n\omega_0 t + \phi_n)
 \end{aligned} \tag{3.15}$$

with c_n and ϕ_n defined by Eqs (3.13) and (3.14).

3.2.3 Evaluation of Fourier Series Coefficients

In this section we will derive Eqs (3.7), (3.8), and (3.9). The evaluation of a_0 , a_n , and b_n coefficients in Eq. (3.6) is accomplished by using simple integral equations, which may be derived by using the orthogonality property of the set of functions involved, namely $\cos(n\omega_0 t)$ and $\sin(n\omega_0 t)$ with integer values for n and m . These functions are orthogonal over the interval from t_0 to $t_0 + T$ for any t_0 . We will often use the value $t_0 = 0$ or $t_0 = -T/2$, but with the understanding that any period may be used. First observe that

$$\int_0^T \sin(m\omega_0 t) dt = 0, \quad \text{for all } m \tag{3.16}$$

and

$$\int_0^T \cos(n\omega_0 t) dt = 0, \quad \text{for all } n \neq 0 \tag{3.17}$$

since the average value of a sinusoid over m and n complete cycles in the period T is zero. The following three cross product terms are also zero for the stated relationships of m and n :

$$\int_0^T \sin(m\omega_0 t) \cos(n\omega_0 t) dt = 0, \quad \text{for all } m, n \tag{3.18}$$

$$\int_0^T \sin(m\omega_0 t) \sin(n\omega_0 t) dt = \begin{cases} 0, & m \neq n \\ \frac{T}{2}, & m = n \end{cases} \tag{3.19}$$

and

$$\int_0^T \cos(m\omega_0 t) \cos(n\omega_0 t) dt = \begin{cases} 0, & m \neq n \\ \frac{T}{2}, & m = n \end{cases} \tag{3.20}$$

Case I Proof of Eq. (3.7), i.e., $a_0 = \frac{1}{T} \int_0^T x(t) dt$

From Eq. (3.6), we have

$$x(t) = a_0 + [a_1 \cos(\omega_0 t) + b_1 \sin(\omega_0 t)] + [a_2 \cos(2\omega_0 t) + b_2 \sin(2\omega_0 t)] + \dots \\ + [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)] + \dots$$

Integrating the above equation over the interval of time from 0 to T gives

$$\int_0^T x(t) dt = \int_0^T a_0 dt + \int_0^T a_1 \cos(\omega_0 t) dt + \int_0^T b_1 \sin(\omega_0 t) dt + \int_0^T a_2 \cos(2\omega_0 t) dt \\ + \int_0^T b_2 \sin(2\omega_0 t) dt + \dots + \int_0^T a_n \cos(n\omega_0 t) dt + \int_0^T b_n \sin(n\omega_0 t) dt + \dots$$

Using the results of Eqs (3.16) and (3.17), all the terms on the RHS of the above equation are found to have a zero value, except the first term. Therefore,

$$\int_0^T x(t) dt = \int_0^T a_0 dt + 0 + 0 + \dots$$

$$\int_0^T x(t) dt = a_0 T$$

$$a_0 = \frac{1}{T} \int_0^T x(t) dt$$

The coefficient a_0 is simply the average value of $x(t)$ over a period, sometimes also known as the dc value of the signal.

Case II Proof of Eq. (3.8), i.e., $a_n = \frac{2}{T} \int_0^T x(t) \cos(n\omega_0 t) dt$

From Eq. (3.6), we have

$$x(t) = a_0 + [a_1 \cos(\omega_0 t) + b_1 \sin(\omega_0 t)] + [a_2 \cos(2\omega_0 t) + b_2 \sin(2\omega_0 t)] + \dots \\ + [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)] + \dots$$

By multiplying both sides of the above equation by $\cos(n\omega_0 t)$ and integrating over the interval of time from 0 to T , we get

$$\begin{aligned} \int_0^T x(t) \cos(n\omega_0 t) dt &= \int_0^T a_0 \cos(n\omega_0 t) dt + \int_0^T a_1 \cos(\omega_0 t) \cos(n\omega_0 t) dt \\ &+ \int_0^T b_1 \sin(\omega_0 t) \cos(n\omega_0 t) dt + \int_0^T a_2 \cos(2\omega_0 t) \cos(n\omega_0 t) dt \\ &+ \int_0^T b_2 \sin(2\omega_0 t) \cos(n\omega_0 t) dt + \dots \\ &+ \int_0^T a_n \cos^2(n\omega_0 t) dt + \int_0^T b_n \sin(n\omega_0 t) \cos(n\omega_0 t) dt + \dots \end{aligned}$$

Using the results of Eqs (3.17), (3.18), and (3.20), all the terms on the RHS of the above equation are found to have a zero value, except the one involving the integration of $\cos^2(n\omega_0 t)$, which has the value $T/2$. Therefore,

$$\int_0^T x(t) \cos(n\omega_0 t) dt = a_n \frac{T}{2}$$

$$a_n = \frac{2}{T} \int_0^T x(t) \cos(n\omega_0 t) dt$$

Case III Proof of Eq. (3.9), i.e., $b_n = \frac{2}{T} \int_0^T x(t) \sin(n\omega_0 t) dt$

From Eq.(3.6), we have

$$\begin{aligned} x(t) &= a_0 + [a_1 \cos(\omega_0 t) + b_1 \sin(\omega_0 t)] + [a_2 \cos(2\omega_0 t) + b_2 \sin(2\omega_0 t)] + \dots \\ &+ [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)] + \dots \end{aligned}$$

By multiplying both sides of the above equation by $\sin(n\omega_0 t)$ and integrating over the interval of time from 0 to T , we get

$$\begin{aligned} \int_0^T x(t) \sin(n\omega_0 t) dt &= \int_0^T a_0 \sin(n\omega_0 t) dt + \int_0^T a_1 \sin(\omega_0 t) \sin(n\omega_0 t) dt \\ &+ \int_0^T b_1 \sin(\omega_0 t) \sin(n\omega_0 t) dt \end{aligned}$$

$$\begin{aligned}
& + \int_0^T a_2 \cos(2\omega_0 t) \sin(n\omega_0 t) dt + \int_0^T b_2 \sin(2\omega_0 t) \sin(n\omega_0 t) dt + \dots \\
& + \int_0^T a_n \cos(n\omega_0 t) \sin(n\omega_0 t) dt + \int_0^T b_n \sin^2(n\omega_0 t) dt + \dots
\end{aligned}$$

Using the results of Eqs (3.16), (3.18), and (3.19), all the terms on the RHS of the above equation are found to have a zero value, except the one involving the integration of $\sin^2(n\omega_0 t)$, which has the value $T/2$. Therefore,

$$\int_0^T x(t) \sin(n\omega_0 t) dt = b_n \frac{T}{2}$$

$$b_n = \frac{2}{T} \int_0^T x(t) \sin(n\omega_0 t) dt$$

Example 3.4 Find the trigonometric Fourier series for the waveform shown in Fig. 3.3.

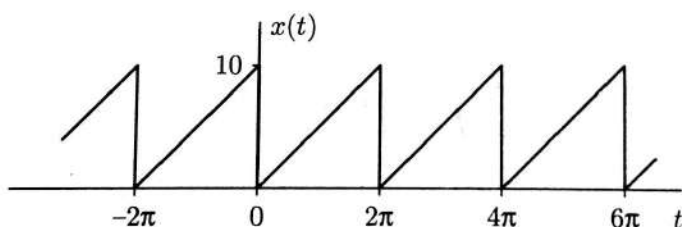


Fig. 3.3 A periodic signal

Solution

The waveform is periodic with period $T = 2\pi$ and fundamental frequency $\omega_0 = 2\pi/T = 1$. The given waveform for one period may be written as

$$x(t) = \frac{10}{2\pi} t, \quad 0 < t < 2\pi$$

From Eq. (3.6), the trigonometric Fourier series of $x(t)$ is

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

where

$$\begin{aligned}
 a_0 &= \frac{1}{T} \int_0^T x(t) dt = \frac{1}{2\pi} \int_0^{2\pi} \frac{10}{2\pi} t dt = \frac{10}{(2\pi)^2} \left[\frac{t^2}{2} \right]_0^{2\pi} = 5 \\
 a_n &= \frac{2}{T} \int_0^T x(t) \cos(n\omega_0 t) dt = \frac{2}{2\pi} \int_0^{2\pi} \frac{10}{2\pi} t \cos(nt) dt \\
 &= \frac{10}{2\pi^2} \left[\frac{t}{n} \sin(nt) + \frac{1}{n^2} \cos(nt) \right]_0^{2\pi} = \frac{10}{2n^2\pi^2} [\cos(2n\pi) - \cos(0)] = 0
 \end{aligned}$$

and

$$\begin{aligned}
 b_n &= \frac{2}{T} \int_0^T x(t) \sin(n\omega_0 t) dt = \frac{2}{2\pi} \int_0^{2\pi} \frac{10}{2\pi} t \sin(nt) dt \\
 &= \frac{10}{2\pi^2} \left[-\frac{t}{n} \cos(nt) + \frac{1}{n^2} \sin(nt) \right]_0^{2\pi} = -\frac{10}{n\pi}
 \end{aligned}$$

Substituting these coefficients (a_0 , a_n , and b_n) into Eq. (3.6), we get

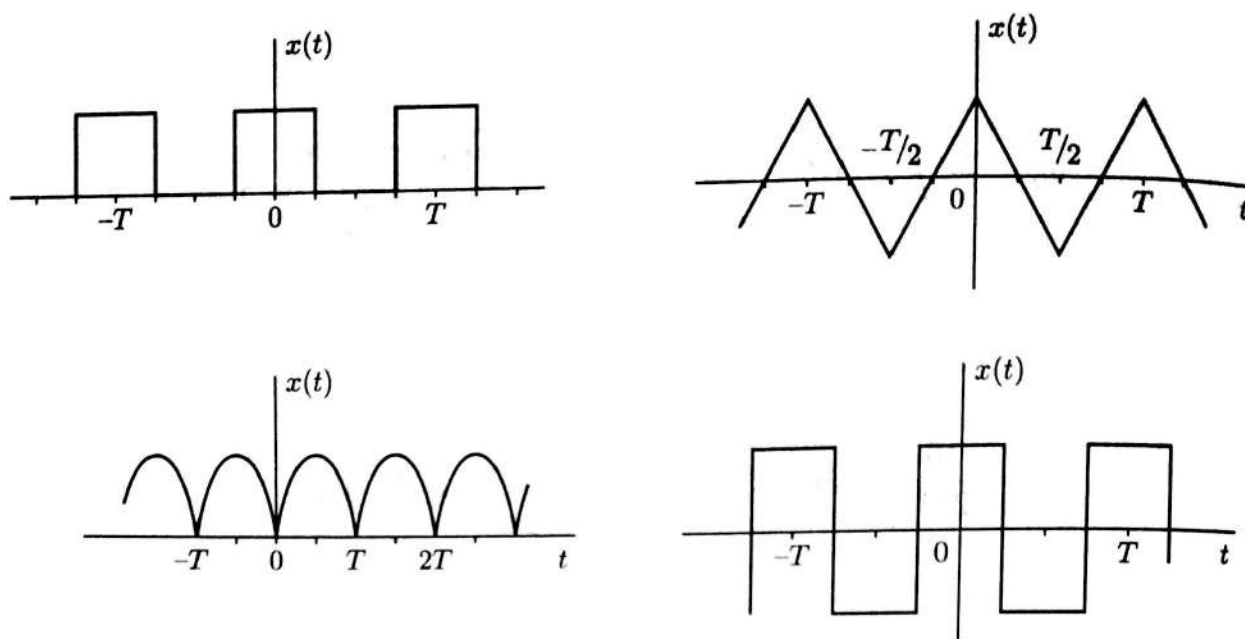
$$\begin{aligned}
 x(t) &= 5 - \sum_{n=1}^{\infty} \left[0 + \frac{10}{n\pi} \sin(nt) \right] \\
 &= 5 - \frac{10}{\pi} \sin(t) - \frac{10}{2\pi} \sin(2t) - \frac{10}{3\pi} \sin(3t) - \dots
 \end{aligned}$$

3.2.4 Symmetry Conditions

The series obtained in Example 3.4 contained only sine terms in addition to a constant term. Other waveforms will have only cosine terms; and sometimes only odd harmonics are present in the series, whether the series contains sine, cosine, or both types of terms. This is the result of certain types of symmetry exhibited by the waveform. Knowledge of such symmetry results in reduced calculations in determining the Fourier series.

1. $x(t)$ is an even signal, i.e., $x(t) = x(-t)$

The signal $x(t) = 2 + t^2 + t^4$ is an example of even signal since the functional values for t and $-t$ are equal. Even signals of special interest to us are the functions $\cos(n\omega_0 t)$. The sum or product of two or more even signals is an even signal, and with the addition of a constant, the even nature of the signal is still preserved. In Fig. 3.4, the waveforms shown represent even functions of t . They are symmetrical w.r.t. the vertical axis.

**Fig. 3.4** Waveforms with even symmetry

If $x(t)$ has an even symmetry, then the Fourier series coefficients a_0 , a_n , and b_n are defined as (we choose our interval for integration from $-T/2$ to $T/2$)

$$\begin{aligned}
 a_0 &= \frac{2}{T} \int_0^{T/2} x(t) dt \\
 a_n &= \frac{4}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt \\
 b_n &= 0
 \end{aligned} \tag{3.21}$$

The Fourier series expansion of an even periodic signal contains only cosine terms and a constant.

Proof

(a) From Eq. (3.7), we have

$$\begin{aligned}
 a_0 &= \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt \\
 &= \frac{1}{T} \left(\int_{-T/2}^0 x(t) dt + \int_0^{T/2} x(t) dt \right) \\
 &= \frac{1}{T} \left(\int_0^{T/2} x(-t) dt + \int_0^{T/2} x(t) dt \right)
 \end{aligned}$$

Since, $x(t)$ is an even signal, i.e., $x(t) = x(-t)$, we have

$$a_0 = \frac{1}{T} \left(\int_0^{T/2} x(t) dt + \int_0^{T/2} x(t) dt \right)$$

$$a_0 = \frac{2}{T} \int_0^{T/2} x(t) dt$$

(b) Now consider Eq. (3.8)

$$a_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cos(n\omega_0 t) dt$$

$$= \frac{2}{T} \left(\int_{-T/2}^0 x(t) \cos(n\omega_0 t) dt + \int_0^{T/2} x(t) \cos(n\omega_0 t) dt \right)$$

$$= \frac{2}{T} \left(\int_0^{T/2} x(-t) \cos(-n\omega_0 t) dt + \int_0^{T/2} x(t) \cos(n\omega_0 t) dt \right)$$

Both $x(t)$ and $\cos(n\omega_0 t)$ are even signals. The product of two even signals results in an even signal, and therefore

$$a_n = \frac{2}{T} \left(\int_0^{T/2} x(t) \cos(n\omega_0 t) dt + \int_0^{T/2} x(t) \cos(n\omega_0 t) dt \right)$$

$$a_n = \frac{4}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt$$

(c) From Eq. (3.9), we have

$$b_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \sin(n\omega_0 t) dt$$

$$= \frac{2}{T} \left(\int_{-T/2}^0 x(t) \sin(n\omega_0 t) dt + \int_0^{T/2} x(t) \sin(n\omega_0 t) dt \right)$$

$$= \frac{2}{T} \left(\int_0^{T/2} x(-t) \sin(-n\omega_0 t) dt + \int_0^{T/2} x(t) \sin(n\omega_0 t) dt \right)$$

$x(t)$ is an even signal and $\sin(n\omega_0 t)$ is an odd signal. The product of an odd and an even signal results in an odd signal, and therefore

$$b_n = \frac{2}{T} \left(- \int_0^{T/2} x(t) \sin(n\omega_0 t) dt + \int_0^{T/2} x(t) \sin(n\omega_0 t) dt \right)$$

$$b_n = 0$$

2. $x(t)$ is an odd signal, i.e., $x(t) = -x(-t)$

The signal $x(t) = t + t^3 + t^5$ is an example of an odd signal since the values of the signal for t and $-t$ are of opposite sign. Odd signals of special interest to us are the functions $\sin(n\omega_0 t)$. The sum of two or more odd signals is an odd signal, but the addition of a constant removes the odd nature of the function. The product of two odd signals is an even signal. The waveforms shown in Fig. 3.5 represent odd functions of t .

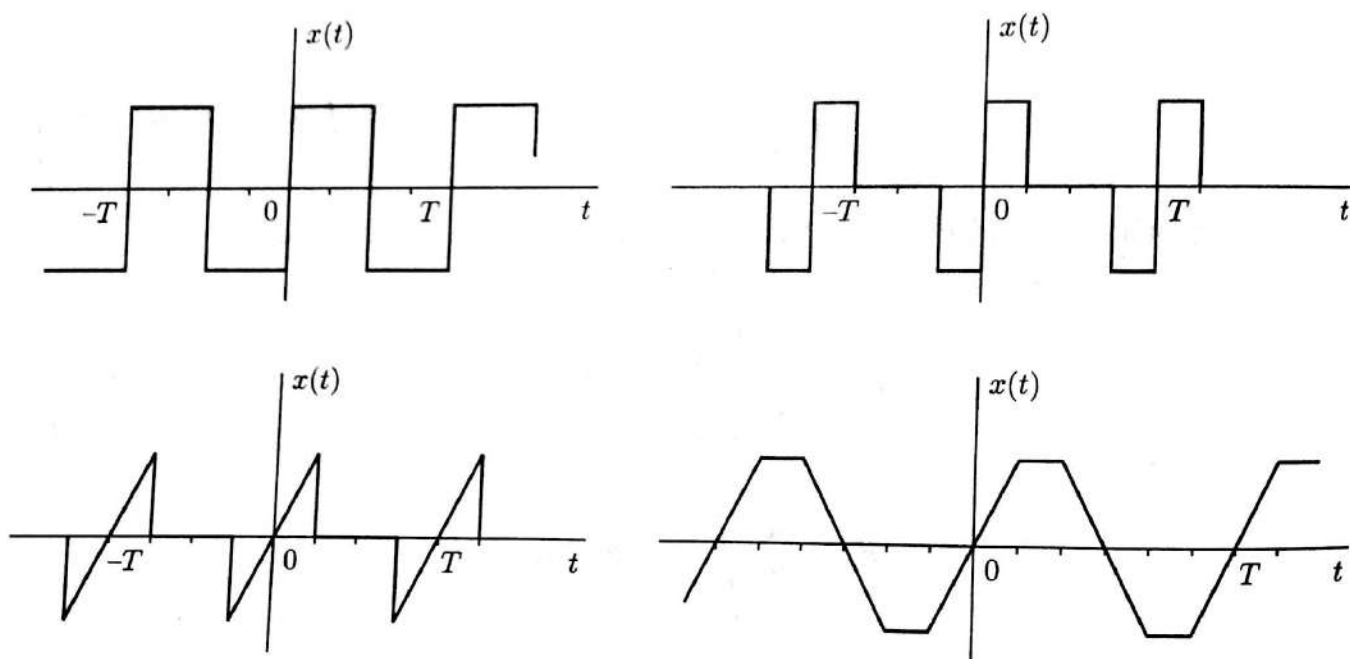


Fig. 3.5 Waveforms with odd symmetry

If $x(t)$ has an odd symmetry, then the Fourier series coefficients a_0 , a_n , and b_n are defined as

$$a_0 = 0$$

$$a_n = 0$$

(3.22)

$$b_n = \frac{4}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt$$

The Fourier series expansion of an odd periodic signal contains only sine terms.

Proof

(a) From Eq. (3.7), we have

$$\begin{aligned}
 a_0 &= \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt \\
 &= \frac{1}{T} \left(\int_{-T/2}^0 x(t) dt + \int_0^{T/2} x(t) dt \right) \\
 &= \frac{1}{T} \left(\int_0^{T/2} x(-t) dt + \int_0^{T/2} x(t) dt \right)
 \end{aligned}$$

Since, $x(t)$ is an odd signal, i.e., $x(t) = -x(-t)$, we have

$$a_0 = \frac{1}{T} \left(- \int_0^{T/2} x(t) dt + \int_0^{T/2} x(t) dt \right) = 0$$

(b) From Eq. (3.8), we have

$$\begin{aligned}
 a_n &= \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cos(n\omega_0 t) dt \\
 &= \frac{2}{T} \left(\int_{-T/2}^0 x(t) \cos(n\omega_0 t) dt + \int_0^{T/2} x(t) \cos(n\omega_0 t) dt \right) \\
 &= \frac{2}{T} \left(\int_0^{T/2} x(-t) \cos(-n\omega_0 t) dt + \int_0^{T/2} x(t) \cos(n\omega_0 t) dt \right)
 \end{aligned}$$

$x(t)$ is an odd signal and $\cos(n\omega_0 t)$ is an even signal. The product of an odd and an even signal results in an odd signal, and therefore

$$a_n = \frac{2}{T} \left(- \int_0^{T/2} x(t) \cos(n\omega_0 t) dt + \int_0^{T/2} x(t) \cos(n\omega_0 t) dt \right) = 0$$

(c) From Eq. (3.9), we have

$$\begin{aligned}
 b_n &= \frac{2}{T} \int_{-T/2}^{T/2} x(t) \sin(n\omega_0 t) dt \\
 &= \frac{2}{T} \left(\int_{-T/2}^0 x(t) \sin(n\omega_0 t) dt + \int_0^{T/2} x(t) \sin(n\omega_0 t) dt \right) \\
 &= \frac{2}{T} \left(\int_0^{T/2} x(-t) \sin(-n\omega_0 t) dt + \int_0^{T/2} x(t) \sin(n\omega_0 t) dt \right)
 \end{aligned}$$

Both $x(t)$ and $\sin(n\omega_0 t)$ are odd signals. The product of two odd signals results in an even signal, and therefore

$$\begin{aligned}
 b_n &= \frac{2}{T} \left(\int_0^{T/2} x(t) \sin(n\omega_0 t) dt + \int_0^{T/2} x(t) \sin(n\omega_0 t) dt \right) \\
 b_n &= \frac{4}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt
 \end{aligned}$$

3. Half-wave symmetry, i.e., $x(t) = -x\left(t \pm \frac{T}{2}\right)$

This symmetry is described by

$$x(t) = -x\left(t \pm \frac{T}{2}\right)$$

and is illustrated in Fig. 3.6, showing that the waveform with t increasing from $-T/2$ to 0 is the negative of the waveform with t increasing from 0 to $T/2$. Clearly, this waveform is neither even nor odd, so it must be both.

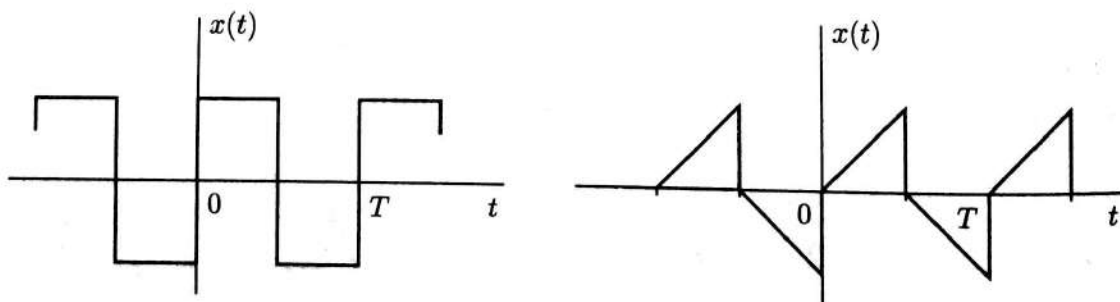


Fig. 3.6 Waveforms with half-wave symmetry

If $x(t)$ has a half-wave symmetry, then the Fourier series coefficients a_0 , a_n , and b_n are defined as

$$\begin{aligned} a_0 &= 0, & a_n &= b_n = 0 \quad n \text{ even} \\ a_n &= \frac{4}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt \quad n \text{ odd} \\ b_n &= \frac{4}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt \quad n \text{ odd} \end{aligned} \quad (3.23)$$

The Fourier series expansion of a periodic signal with half-wave symmetry contains only odd harmonics.

Proof

(a) From Eq. (3.7), we have

$$\begin{aligned} a_0 &= \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt \\ &= \frac{1}{T} \left(\int_{-T/2}^0 x(t) dt + \int_0^{T/2} x(t) dt \right) \\ &= \frac{1}{T} \left(\int_0^{T/2} x(t - T/2) dt + \int_0^{T/2} x(t) dt \right) \end{aligned}$$

$x(t)$ has a half-wave symmetry, $-x(t) = x(t - T/2)$.

$$a_0 = \frac{1}{T} \left(- \int_0^{T/2} x(t) dt + \int_0^{T/2} x(t) dt \right) = 0$$

(b) From Eq. (3.8)

$$\begin{aligned} a_n &= \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cos(n\omega_0 t) dt \\ &= \frac{2}{T} \left(\int_{-T/2}^0 x(t) \cos(n\omega_0 t) dt + \int_0^{T/2} x(t) \cos(n\omega_0 t) dt \right) \end{aligned}$$

$$\begin{aligned}
&= \frac{2}{T} \left(\int_0^{T/2} x(t - T/2) \cos(n\omega_0(t - T/2)) dt + \int_0^{T/2} x(t) \cos(n\omega_0 t) dt \right) \\
&= \frac{2}{T} \left(- \int_0^{T/2} x(t) \cos(n\omega_0(t - T/2)) dt + \int_0^{T/2} x(t) \cos(n\omega_0 t) dt \right)
\end{aligned}$$

Since, $x(t)$ has a half-wave symmetry, i.e., $-x(t) = x(t - T/2)$ and

$$\begin{aligned}
\cos(n\omega_0(t - T/2)) &= \cos\left(n\omega_0 t - n\omega_0 \frac{T}{2}\right) \\
&= \cos\left(n\omega_0 t - n \frac{2\pi}{T} \frac{T}{2}\right) \\
&= \cos(n\omega_0 t - n\pi) \\
&= \begin{cases} \cos(n\omega_0 t), & n \text{ even} \\ -\cos(n\omega_0 t), & n \text{ odd} \end{cases}
\end{aligned}$$

therefore, we have

$$\begin{aligned}
a_n &= \begin{cases} \frac{2}{T} \left(- \int_0^{T/2} x(t) \cos(n\omega_0 t) dt + \int_0^{T/2} x(t) \cos(n\omega_0 t) dt \right), & n \text{ even} \\ \frac{2}{T} \left(\int_0^{T/2} x(t) \cos(n\omega_0 t) dt + \int_0^{T/2} x(t) \cos(n\omega_0 t) dt \right), & n \text{ odd} \end{cases} \\
a_n &= \begin{cases} 0, & n \text{ even} \\ \frac{4}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt, & n \text{ odd} \end{cases}
\end{aligned}$$

(c) From Eq. (3.9)

$$\begin{aligned}
b_n &= \frac{2}{T} \int_{-T/2}^{T/2} x(t) \sin(n\omega_0 t) dt \\
&= \frac{2}{T} \left(\int_{-T/2}^0 x(t) \sin(n\omega_0 t) dt + \int_0^{T/2} x(t) \sin(n\omega_0 t) dt \right)
\end{aligned}$$

$$\begin{aligned}
&= \frac{2}{T} \left(\int_0^{T/2} x(t - T/2) \sin(n\omega_0(t - T/2)) dt + \int_0^{T/2} x(t) \sin(n\omega_0 t) dt \right) \\
&= \frac{2}{T} \left(- \int_0^{T/2} x(t) \sin(n\omega_0(t - T/2)) dt + \int_0^{T/2} x(t) \sin(n\omega_0 t) dt \right)
\end{aligned}$$

Since, $x(t)$ has a half-wave symmetry, i.e., $-x(t) = x(t - T/2)$ and

$$\begin{aligned}
\sin(n\omega_0(t - T/2)) &= \sin\left(n\omega_0 t - n\omega_0 \frac{T}{2}\right) \\
&= \sin\left(n\omega_0 t - n \frac{2\pi}{T} \frac{T}{2}\right) \\
&= \sin(n\omega_0 t - n\pi) \\
&= \begin{cases} \sin(n\omega_0 t), & n \text{ even} \\ -\sin(n\omega_0 t), & n \text{ odd} \end{cases}
\end{aligned}$$

Exclusive @sTc

therefore, we have

$$\begin{aligned}
b_n &= \begin{cases} \frac{2}{T} \left(- \int_0^{T/2} x(t) \sin(n\omega_0 t) dt + \int_0^{T/2} x(t) \sin(n\omega_0 t) dt \right), & n \text{ even} \\ \frac{2}{T} \left(\int_0^{T/2} x(t) \sin(n\omega_0 t) dt + \int_0^{T/2} x(t) \sin(n\omega_0 t) dt \right), & n \text{ odd} \end{cases} \\
b_n &= \begin{cases} 0, & n \text{ even} \\ \frac{4}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt, & n \text{ odd} \end{cases}
\end{aligned}$$

4. Quarter-wave symmetry

A periodic function possesses a quarter-wave symmetry if

- (i) it has either odd or even symmetry and
- (ii) it has half-wave symmetry.

A waveform that satisfies the above conditions and if it is shifted towards right or left by $T/4$ period gives a new waveform with a half-wave symmetry, but the even (odd) symmetry of the original unshifted waveform has changed into odd (even) symmetry, is said to be having a quarter-wave symmetry.

- (a) If a periodic function is even, and has quarter-wave symmetry, the Fourier coefficients are

$$\begin{aligned} a_0 &= 0, \quad b_n = 0, \quad \text{for all } n \\ a_n &= 0, \quad \text{for even } n \end{aligned} \quad (3.24)$$

$$a_n = \frac{8}{T} \int_0^{T/4} x(t) \cos(n\omega_0 t) dt, \quad \text{for odd } n$$

- (b) If a periodic function is odd, and has quarter-wave symmetry, the Fourier coefficients are

$$\begin{aligned} a_0 &= 0, \quad a_n = 0, \quad \text{for all } n \\ b_n &= 0, \quad \text{for even } n \end{aligned} \quad (3.25)$$

$$b_n = \frac{8}{T} \int_0^{T/4} x(t) \sin(n\omega_0 t) dt, \quad \text{for odd } n$$

Table 3.1 summarizes the symmetry conditions.

Table 3.1 Fourier Series Representation

Name of symmetry	Condition	Property	a_0	a_n ($n \neq 0$)	b_n
Even	$x(t) = x(-t)$	cosine terms only	$\frac{4}{T} \int_0^{T/2} x(t) dt$	$\frac{4}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt$	0
Odd	$x(t) = -x(-t)$	sine terms only	0	0	$\frac{4}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt$
Half-wave	$x(t) = -x(t \pm T/2)$	odd n only	0	$\frac{4}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt$	$\frac{4}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt$

Example 3.5 Find the trigonometric Fourier series for the square wave shown in Fig. 3.7 and plot the line spectrum.

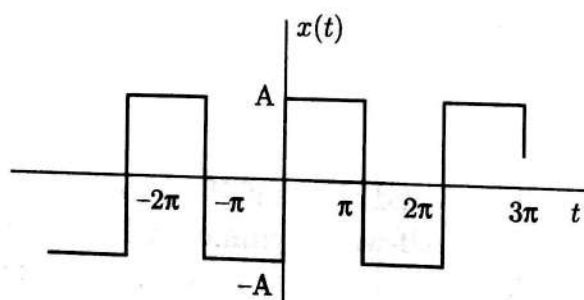


Fig. 3.7 Square wave

Solution

The waveform is periodic with period $T = 2\pi$ and fundamental frequency $\omega_0 = 2\pi/T = 1$. The given waveform for one period may be written as

$$x(t) = \begin{cases} A, & 0 < t < \pi \\ -A, & \pi < t < 2\pi \end{cases}$$

The wave is an odd function, so the symmetry condition requires that [Eq. (3.22)]

$$a_0 = 0, \quad a_n = 0$$

$$\begin{aligned} b_n &= \frac{4}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt = \frac{4}{2\pi} \int_0^{\pi} A \sin(nt) dt = \frac{2A}{\pi} \left[-\frac{1}{n} \cos(nt) \right]_0^{\pi} \\ &= \frac{2A}{n\pi} (1 - \cos(n\pi)) \\ b_n &= \begin{cases} \frac{4A}{n\pi}, & n = 1, 3, 5, \dots \\ 0, & n = 2, 4, 6, \dots \end{cases} \end{aligned}$$

The trigonometric Fourier series for the square wave is

$$\begin{aligned} x(t) &= \sum_{n=1}^{\infty} b_n \sin(nt) \\ &= \frac{4A}{\pi} \sin(t) + \frac{4A}{3\pi} \sin(3t) + \frac{4A}{5\pi} \sin(5t) + \dots \end{aligned}$$

The series contains only odd-harmonic sine terms. Since the wave in Fig. 3.7 is odd, its series contains only sine terms, and since it also has a half-wave symmetry, only odd harmonics are present.

The line spectrum for this series is shown in Fig. 3.8. From Eq.(3.13), we have

$$c_n = \sqrt{a_n^2 + b_n^2}$$

$$c_n = \sqrt{0 + b_n^2} = |b_n|$$

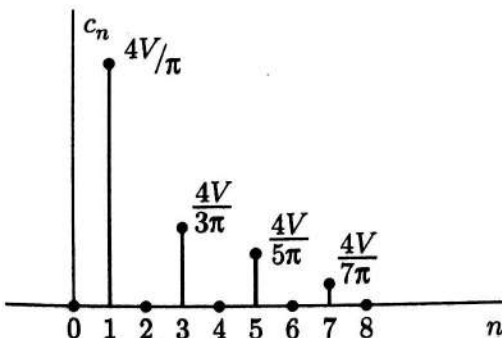


Fig. 3.8 Line spectrum

Example 3.6 Find the trigonometric Fourier series for the triangular wave shown in Fig. 3.9 and plot the line spectrum.

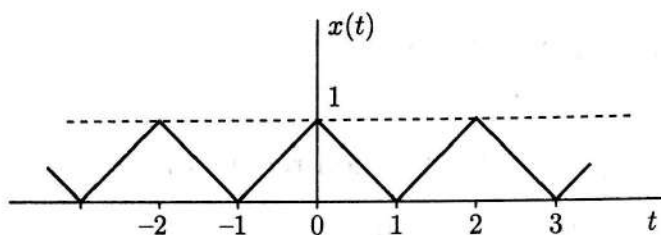


Fig. 3.9 Triangular wave

Solution

The waveform is periodic with period $T = 2$ and fundamental frequency $\omega_0 = 2\pi/T = \pi$. The given waveform for one period may be written as

$$x(t) = \begin{cases} 1 - t, & 0 < t < 1 \\ t - 1, & 1 < t < 2 \end{cases}$$

The wave is an even function, so the symmetry condition requires that [Eq. (3.21)] $b_n = 0$,

$$a_0 = \frac{2}{T} \int_0^{T/2} x(t) dt = \frac{2}{2} \int_0^1 (1 - t) dt = \left[t - \frac{t^2}{2} \right]_0^1 = \frac{1}{2}$$

and

$$\begin{aligned} a_n &= \frac{4}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt = \frac{4}{2} \int_0^1 (1 - t) \cos(n\pi t) dt \\ &= 2 \left(\left[(1 - t) \frac{\sin(n\pi t)}{n\pi} \right]_0^1 - \left[\frac{\cos(n\pi t)}{n^2\pi^2} \right]_0^1 \right) \\ &= \frac{2}{n^2\pi^2} (1 - \cos(n\pi)) \\ a_n &= \begin{cases} \frac{4}{n^2\pi^2}, & n = 1, 3, 5, \dots \\ 0, & n = 2, 4, 6, \dots \end{cases} \end{aligned}$$

Substituting the values of a_0 , a_n , and b_n into Eq. (3.6), gives

$$\begin{aligned} x(t) &= a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) \\ &= \frac{1}{2} + \frac{4}{\pi^2} \cos(\pi t) + \frac{4}{(3\pi)^2} \cos(3\pi t) + \frac{4}{(5\pi)^2} \cos(5\pi t) + \dots \end{aligned}$$

The series contains only odd-harmonic cosine terms. The coefficients decrease as $1/n^2$, and thus the series converges rapidly. This fact is evident from the line spectrum shown in Fig. 3.10. From Eq. (3.10), we have

$$c_n = \sqrt{a_n^2 + b_n^2}$$

$$c_n = \sqrt{a_n^2 + 0} = |a_n|$$

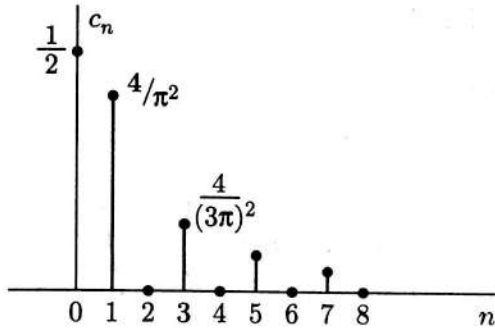


Fig. 3.10 Line spectrum

Example 3.7 Find the trigonometric Fourier series for the waveform shown in Fig. 3.11 and sketch the line spectrum.

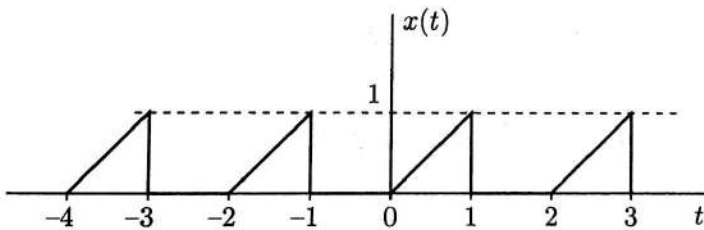


Fig. 3.11

Solution

The waveform is periodic with period $T = 2$ and fundamental frequency $\omega_0 = 2\pi/T = \pi$. The given waveform for one period may be written as

$$x(t) = \begin{cases} t, & 0 < t < 1 \\ 0, & 1 < t < 2 \end{cases}$$

Since the wave is neither even nor odd, the series will contain both sine and cosine terms.

$$a_0 = \frac{1}{T} \int_0^T x(t) dt = \frac{1}{2} \left(\int_0^1 t dt + \int_1^2 0 dt \right) = \frac{1}{4}$$

Now we determine a_n and b_n :

$$\begin{aligned}
 a_n &= \frac{2}{T} \int_0^T x(t) \cos(n\omega_0 t) dt = \frac{2}{2} \left(\int_0^1 t \cos(n\pi t) dt + \int_1^2 0 \cos(n\pi t) dt \right) = \int_0^1 t \cos(n\pi t) dt + 0 \\
 &= \left[t \frac{\sin(n\pi t)}{n\pi} \right]_0^1 + \left[\frac{\cos(n\pi t)}{(n\pi)^2} \right]_0^1 = \frac{1}{(n\pi)^2} (\cos(n\pi) - 1) \\
 a_n &= \begin{cases} -\frac{2}{(n\pi)^2}, & n = 1, 3, 5, \dots \\ 0, & n = 2, 4, 6, \dots \end{cases}
 \end{aligned}$$

$$\begin{aligned}
 b_n &= \frac{2}{T} \int_0^T x(t) \sin(n\omega_0 t) dt = \frac{2}{2} \left(\int_0^1 t \sin(n\pi t) dt + \int_1^2 0 \sin(n\pi t) dt \right) = \int_0^1 t \sin(n\pi t) dt + 0 \\
 &= \left[-t \frac{\cos(n\pi t)}{n\pi} \right]_0^1 + \left[\frac{\sin(n\pi t)}{(n\pi)^2} \right]_0^1 = -\frac{1}{n\pi} \cos(n\pi) \\
 b_n &= \begin{cases} \frac{1}{n\pi}, & n = 1, 3, 5, \dots \\ -\frac{1}{n\pi}, & n = 2, 4, 6, \dots \end{cases}
 \end{aligned}$$

Substituting the values of a_0 , a_n , and b_n into Eq. (3.6), gives

$$\begin{aligned}
 x(t) &= a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)] \\
 &= \frac{1}{4} - \frac{2}{\pi^2} \cos(\pi t) - \frac{2}{(3\pi)^2} \cos(3\pi t) - \frac{2}{(5\pi)^2} \cos(5\pi t) - \dots \\
 &\quad + \frac{1}{\pi} \sin(\pi t) - \frac{1}{2\pi} \sin(2\pi t) + \frac{1}{3\pi} \sin(3\pi t) - \dots
 \end{aligned}$$

The line spectrum is shown in Fig. 3.12. The even harmonic amplitudes are given directly by $|b_n|$ since there are no even-harmonic cosine terms. However, the odd-harmonic amplitudes must be computed using $c_n = \sqrt{a_n^2 + b_n^2}$, $n \geq 1$ and $c_0 = a_0$. Thus,

$$c_1 = \sqrt{\left(\frac{2}{\pi}\right)^2 + \left(\frac{1}{\pi}\right)^2} = 0.377$$

$$c_3 = 0.109$$

$$c_5 = 0.064$$

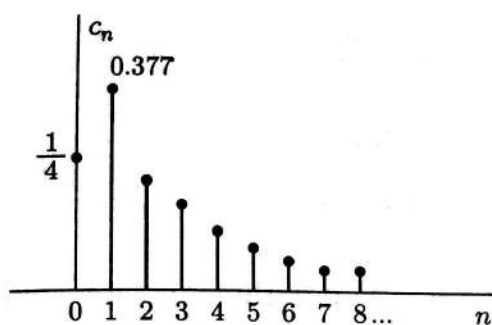


Fig. 3.12 Line spectrum

Example 3.8 Find the trigonometric Fourier series for the half-wave rectified sine wave shown in Fig. 3.13 and sketch the line spectrum.

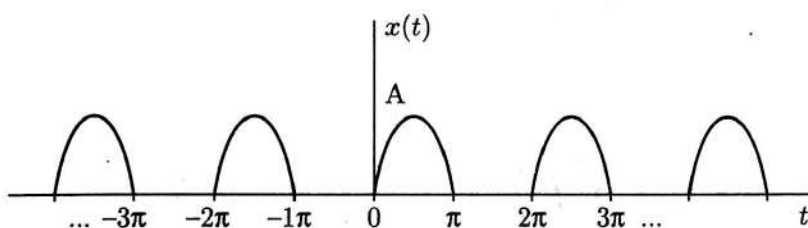


Fig. 3.13 Half-wave rectified sine wave

Solution

The waveform is periodic with period $T = 2\pi$ and fundamental frequency $\omega_0 = 2\pi/T = 1$. The given waveform for one period may be written as

$$x(t) = \begin{cases} A \sin(t), & 0 < t < \pi \\ 0, & \pi < t < 2\pi \end{cases}$$

The wave shows no symmetry and we therefore expect the series to contain both sine and cosine terms.

$$a_0 = \frac{1}{T} \int_0^T x(t) dt = \frac{1}{2\pi} \left(\int_0^{\pi} A \sin(t) dt + \int_{\pi}^{2\pi} 0 dt \right) = \frac{A}{2\pi} [-\cos(t)]_0^{\pi} = \frac{A}{\pi}$$

Now we determine a_n :

$$\begin{aligned} a_n &= \frac{2}{T} \int_0^T x(t) \cos(n\omega_0 t) dt = \frac{2}{2\pi} \left(\int_0^{\pi} A \sin(t) \cos(nt) dt + \int_{\pi}^{2\pi} 0 \times \cos(nt) dt \right) \\ &= \frac{A}{\pi} \int_0^{\pi} \sin(t) \cos(nt) dt + 0 = \frac{A}{\pi} \left[\frac{-n \sin(t) \sin(nt) - \cos(nt) \cos(t)}{1 - n^2} \right]_0^{\pi} \end{aligned}$$

$$a_n = \frac{A}{\pi(1-n^2)} (\cos(n\pi) + 1)$$

$$a_n = \begin{cases} 0, & n = 3, 5, \dots \\ \frac{2A}{\pi(1-n^2)}, & n = 2, 4, 6, \dots \end{cases}$$

However, this expression is indeterminate for $n = 1$ and therefore we must integrate separately for a_1 :

$$\begin{aligned} a_1 &= \frac{2}{T} \int_0^T x(t) \cos(\omega_0 t) dt = \frac{2}{2\pi} \left(\int_0^\pi A \sin(t) \cos(t) dt + \int_\pi^{2\pi} 0 \cos(t) dt \right) \\ &= \frac{A}{2\pi} \int_0^\pi \sin(2t) dt + 0 = \frac{A}{2\pi} \left[-\frac{\cos(2t)}{2} \right]_0^\pi = 0 \end{aligned}$$

Now we evaluate b_n :

$$\begin{aligned} b_n &= \frac{2}{T} \int_0^T x(t) \sin(n\omega_0 t) dt = \frac{2}{2\pi} \left(\int_0^\pi A \sin(t) \sin(nt) dt + \int_\pi^{2\pi} 0 \sin(nt) dt \right) \\ &= \frac{A}{\pi} \int_0^\pi \sin(t) \sin(nt) dt + 0 = \frac{A}{\pi} \left[\frac{n \sin(t) \cos(nt) - \sin(nt) \cos(t)}{1-n^2} \right]_0^\pi = 0 \end{aligned}$$

This expression is also indeterminate for $n = 1$, and b_1 is evaluated separately:

$$\begin{aligned} b_1 &= \frac{2}{T} \int_0^T x(t) \sin(\omega_0 t) dt = \frac{2}{2\pi} \left(\int_0^\pi A \sin^2(t) dt + \int_\pi^{2\pi} 0 \sin(t) dt \right) \\ &= \frac{A}{\pi} \int_0^\pi \sin^2(t) dt = \frac{A}{2\pi} \left[t - \frac{\sin(2t)}{2} \right]_0^\pi = \frac{A}{2} \end{aligned}$$

Substituting the values of a_0 , a_n , and b_n into Eq. (3.6) gives

$$x(t) = \frac{A}{\pi} \left(1 + \frac{\pi}{2} \sin(t) - \frac{2}{3} \cos(2t) - \frac{2}{15} \cos(4t) - \dots \right)$$

The line spectrum, shown in Fig. 3.14, is determined using $c_n = \sqrt{a_n^2 + b_n^2}$ for $n \geq 1$ and $c_0 = a_0$.

$$c_0 = \frac{A}{\pi}, \quad c_1 = \sqrt{a_1^2 + b_1^2} = \frac{A}{2}, \quad c_n = |a_n| \quad n \geq 2$$

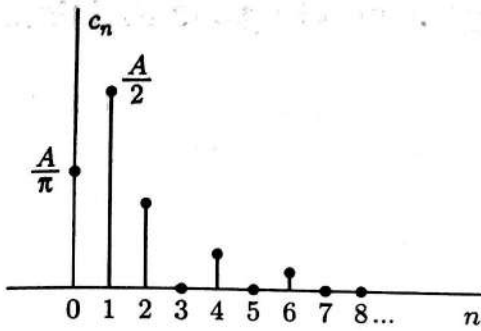


Fig. 3.14 Line spectrum

3.2.5 Gibbs Phenomenon

The Gibbs phenomenon is present only when there is a jump discontinuity in $x(t)$. When a continuous function $x(t)$ is synthesized by using the first N terms of the Fourier series, the synthesized function approaches $x(t)$ for all t as $N \rightarrow \infty$. No Gibbs phenomenon appears.

The Gibbs phenomenon occurs near points of discontinuity of a periodic signal $x(t)$ that is approximated by a Fourier series in which only a finite number of terms are kept. When we use only the first N terms in the Fourier series to synthesize a signal, we are abruptly terminating the series, giving a unit weight to the first N harmonics and zero weight to all the remaining harmonics beyond N . This abrupt termination of the series causes the Gibbs phenomenon in synthesis of discontinuous functions. Near a point of discontinuity, the Fourier series approximation oscillates about the numerical value it should achieve according to the Fourier convergence theorem, which is valid in the infinite series limit. Further, the overshoot near the discontinuity does not vanish as more and more modes are retained. Instead, the overshoot is finite no matter what finite number of modes N are retained, even though the region of overshoot gets progressively smaller as $N \rightarrow \infty$.

3.2.6 Exponential Fourier Series

The exponential Fourier series is expressed as

$$x(t) = \sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t} \quad (3.26)$$

where

$$X_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt \quad (3.27)$$

are the exponential Fourier series coefficients of the signal $x(t)$. The integrations can be carried out from $-T/2$ to $T/2$, or over any other full period that might simplify the calculation. Equation (3.26) is referred to as the *synthesis* equation and Eq. (3.27) as the *analysis* equation.

The Fourier series coefficients are known as a *frequency domain representation* of $x(t)$ because each Fourier series coefficient is associated with a complex sinusoid of a different frequency.

3.2.7 Relationship Between Trigonometric and Exponential Fourier Series

If we express each of the sine and cosine terms in the trigonometric series [Eq. (3.6)] by its exponential equivalent, the result is a series of exponential terms:

$$\begin{aligned}
 x(t) &= a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)] \\
 &= a_0 + \sum_{n=1}^{\infty} \left[a_n \frac{e^{jn\omega_0 t} + e^{-jn\omega_0 t}}{2} + b_n \frac{e^{jn\omega_0 t} - e^{-jn\omega_0 t}}{2j} \right] \\
 &= a_0 + \sum_{n=1}^{\infty} \left[a_n \frac{e^{jn\omega_0 t} + e^{-jn\omega_0 t}}{2} - jb_n \frac{e^{jn\omega_0 t} - e^{-jn\omega_0 t}}{2} \right] \\
 x(t) &= a_0 + \sum_{n=1}^{\infty} \left[\left(\frac{a_n - jb_n}{2} \right) e^{jn\omega_0 t} + \left(\frac{a_n + jb_n}{2} \right) e^{-jn\omega_0 t} \right] \quad (3.28)
 \end{aligned}$$

Assume

$$X_n = \frac{a_n - jb_n}{2} \quad (3.29)$$

Substituting the values of a_n and b_n from Eqs (3.8) and (3.9) into the above equation gives

$$\begin{aligned}
 X_n &= \frac{1}{2} \left(\frac{2}{T} \int_0^T x(t) \cos(n\omega_0 t) dt - j \frac{2}{T} \int_0^T x(t) \sin(n\omega_0 t) dt \right) \\
 &= \frac{1}{2} \frac{2}{T} \int_0^T x(t) (\cos(n\omega_0 t) - j \sin(n\omega_0 t)) dt \\
 X_n &= \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt
 \end{aligned}$$

Substituting $n = 0$ in the above equation gives

$$X_0 = \frac{1}{T} \int_0^T x(t) dt = a_0$$

$$X_0 = a_0 \quad (3.30)$$

Again, consider

$$\begin{aligned}
 X_n &= \frac{a_n - jb_n}{2} \\
 X_{-n} &= \frac{a_{-n} - jb_{-n}}{2}
 \end{aligned}$$

$$a_n = \frac{2}{T} \int_0^T x(t) \cos(n\omega_0 t) dt$$

$$a_{-n} = \frac{2}{T} \int_0^T x(t) \cos(-n\omega_0 t) dt$$

$$a_{-n} = \frac{2}{T} \int_0^T x(t) \cos(n\omega_0 t) dt = a_n$$

$$a_{-n} = a_n \quad \text{an even Fourier series coefficient} \quad (3.31)$$

and

$$b_n = \frac{2}{T} \int_0^T x(t) \sin(n\omega_0 t) dt$$

$$b_{-n} = \frac{2}{T} \int_0^T x(t) \sin(-n\omega_0 t) dt$$

$$= -\frac{2}{T} \int_0^T x(t) \sin(n\omega_0 t) dt = -b_n$$

$$b_{-n} = -b_n \quad \text{an odd Fourier series coefficient} \quad (3.32)$$

Substituting $a_{-n} = a_n$ and $b_{-n} = -b_n$ in the expression of X_{-n} gives

$$X_{-n} = \frac{a_n + jb_n}{2} \quad (3.33)$$

from Eqs (3.28), (3.29), (3.30), and Eq. (3.33), we have

$$x(t) = X_0 + \sum_{n=1}^{\infty} [X_n e^{jn\omega_0 t} + X_{-n} e^{-jn\omega_0 t}]$$

$$x(t) = \sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t}$$

The exponential Fourier series is just another way of representing trigonometric Fourier series (or vice versa). The two form carry identical information: no more, no less. The reasons for preferring the exponential form are that it is more compact, and the expression for deriving the exponential coefficients is also more compact, than those in the trigonometric series. Furthermore, the LTI system response to exponential signals is also

simpler than the system response to sinusoids. In addition, the exponential form proves to be much easier than the trigonometric form to manipulate mathematically.

A minor disadvantage of the exponential form is that it cannot be visualized as easily as sinusoids. For intuitive and qualitative understanding, the sinusoids have an edge over exponentials. Fortunately, this difficulty can be overcome readily because of the close connection between exponential and Fourier spectra. Using Eqs (3.29), (3.30), and (3.33), we may convert the exponential Fourier series into the trigonometric Fourier series. Note that

$$a_0 = X_0 \quad (3.34)$$

$$a_n = 2 \operatorname{Re}\{X_n\} \quad \text{or} \quad a_n = X_n + X_{-n} \quad (3.35)$$

$$b_n = -2 \operatorname{Im}\{X_n\} \quad \text{or} \quad b_n = j(X_n - X_{-n}) \quad (3.36)$$

3.2.8 Line Spectrum

A plot showing each of the harmonic amplitudes in the wave is called the *line spectrum*. The lines decrease rapidly for waves with rapidly convergent series. Waves with discontinuities, such as the sawtooth and square wave, have spectra with slowly decreasing amplitudes since their series have strong high harmonics. The harmonic content and the line spectrum of a wave are part of the very nature of that wave and never change, regardless of the method of analysis. Shifting the origin gives the trigonometric series a completely different appearance, and the exponential series coefficients also change greatly. However, the same harmonic always appear in the series and their amplitudes remains the same.

From Eq. (3.29), the *magnitude spectrum* is

$$|X_n| = |X_{-n}| = \frac{\sqrt{a_n^2 + b_n^2}}{2} = \frac{c_n}{2} \quad (3.37)$$

The *magnitude spectrum* is an even function.

$$c_n = \begin{cases} 2|X_n|, & n \geq 1 \\ X_0, & n = 0 \end{cases} \quad (3.38)$$

and, the *phase spectrum* is

$$\angle X_n = \tan^{-1} \left(\frac{-b_n}{a_n} \right) = \theta_n \quad (3.39)$$

$$\angle X_{-n} = \tan^{-1} \left(\frac{-b_{-n}}{a_{-n}} \right)$$

$$\begin{aligned} &= \tan^{-1} \left(\frac{b_n}{a_n} \right) = -\tan^{-1} \left(\frac{-b_n}{a_n} \right) = -\theta_n = -\angle X_n \\ &= -\angle X_n \end{aligned} \quad (3.40)$$

The *phase spectrum* is an odd function.

3.2.9 Concept of Negative Frequency

The existence of the spectrum at negative frequencies is somewhat disturbing because, by definition, the frequency is a positive quantity. How do we interpret a negative frequency?

Negative frequencies are present in the exponential Fourier series because the mathematical model of a signal requires the use of negative frequencies. The exponential spectra are a graphical representation of coefficients X_n as a function of ω . Existence of the spectrum at $\omega = -n\omega_0$ is merely an indication that an exponential component $e^{-jn\omega_0 t}$ exists in the series. The function $e^{-jn\omega_0 t}$ of frequency $-n\omega_0$ (negative frequency) represents the counterpart of the signal $e^{jn\omega_0 t}$ of frequency $n\omega_0$ (positive frequency), and when these two exponentials are combined, they provide a real function (sinusoid) of frequency $n\omega_0$.

Example 3.9 Find the exponential Fourier series for the waveform shown in Fig. 3.3. Using the coefficients of this exponential series, obtain a_n and b_n of the trigonometric series.

Solution

The waveform is periodic with period $T = 2\pi$ and fundamental frequency $\omega_0 = 2\pi/T = 1$. The given waveform for one period may be written as

$$x(t) = \frac{10}{2\pi}t, \quad 0 < t < 2\pi$$

From Eq. (3.27), the exponential Fourier series coefficient is

$$\begin{aligned} X_n &= \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt = \frac{1}{2\pi} \int_0^{2\pi} \frac{10}{2\pi} t e^{-jnt} dt \\ &= \frac{10}{(2\pi)^2} \left[-\frac{1}{jn} t e^{-jnt} - \frac{1}{(jn)^2} e^{-jnt} \right]_0^{2\pi} = j \frac{10}{2n\pi} \end{aligned}$$

The average value is

$$X_0 = a_0 = \frac{1}{T} \int_0^T x(t) dt = \frac{1}{2\pi} \int_0^{2\pi} \frac{10}{2\pi} t dt = \frac{10}{(2\pi)^2} \left[\frac{t^2}{2} \right]_0^{2\pi} = 5$$

The exponential Fourier series for the given waveform is

$$\begin{aligned} x(t) &= \sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t} = \sum_{n=-\infty}^{\infty} X_n e^{jnt} \\ &= \dots - j \frac{10}{4\pi} e^{-j2t} - j \frac{10}{2\pi} e^{-jt} + 5 + j \frac{10}{2\pi} e^{jt} + j \frac{10}{4\pi} e^{j2t} + \dots \end{aligned}$$

The trigonometric Fourier series coefficients are [from Eqs (3.34), (3.35), and (3.36)]

$$a_0 = X_0 = 5, \quad a_n = 0, \quad b_n = -\frac{10}{n\pi}$$

and so

$$x(t) = 5 - \frac{10}{\pi} \sin(t) - \frac{10}{2\pi} \sin(2t) - \frac{10}{3\pi} \sin(3t) - \dots$$

Example 3.10 Find the exponential Fourier series for the half-wave rectified sine wave shown in Fig. 3.13 and sketch the line spectrum.

Solution

The waveform is periodic with period $T = 2\pi$ and fundamental frequency $\omega_0 = 2\pi/T = 1$. The given waveform for one period may be written as

$$x(t) = \begin{cases} A \sin(t), & 0 < t < \pi \\ 0, & \pi < t < 2\pi \end{cases}$$

@sTc

From Eq. (3.27), the exponential Fourier series coefficient is

$$\begin{aligned} X_n &= \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt = \frac{1}{2\pi} \left(\int_0^\pi A \sin(t) e^{-jnt} dt + \int_\pi^{2\pi} 0 e^{-jnt} dt \right) \\ &= \frac{A}{2\pi} \int_0^\pi \sin(t) e^{-jnt} dt = \frac{A}{2\pi} I \end{aligned}$$

where

$$I = \int_0^\pi e^{-jnt} \sin(t) dt$$

$$I = [-e^{-jnt} \cos(t)]_0^\pi - jn \int_0^\pi e^{-jnt} \cos(t) dt$$

$$I = [-e^{-jnt} \cos(t)]_0^\pi + [-jn e^{-jnt} \sin(t)]_0^\pi + n^2 \int_0^\pi e^{-jnt} \sin(t) dt$$

$$I = [-e^{-jnt} \cos(t)]_0^\pi + [-jn e^{-jnt} \sin(t)]_0^\pi + n^2 I$$

$$I(1 - n^2) = [e^{-jnt}(-\cos(t) - jn \sin(t))]_0^\pi$$

$$I = \frac{(e^{-jn\pi} + 1)}{(1 - n^2)}$$

Therefore,

$$X_n = \frac{A}{2\pi} I = \frac{A(e^{-jn\pi} + 1)}{2\pi(1 - n^2)} = \begin{cases} \frac{A(e^{-jn\pi} + 1)}{\pi(1 - n^2)}, & n = \pm 2, \pm 4, \pm 6, \dots \\ 0, & n = \pm 3, \pm 5, \pm 7, \dots \end{cases}$$

For $n = \pm 1$, the expression for X_n becomes indeterminate. L'Hospital's rule may be applied, i.e., the numerator and denominator are separately differentiated w.r.t. n , after which n is allowed to approach ± 1 .

$$X_1 = \lim_{n \rightarrow 1} \frac{d/dn A(e^{-jn\pi} + 1)}{d/dn 2\pi(1 - n^2)} = -j \frac{A}{4}$$

Similarly,

$$X_{-1} = \lim_{n \rightarrow -1} \frac{d/dn A(e^{-jn\pi} + 1)}{d/dn 2\pi(1 - n^2)} = j \frac{A}{4}$$

The average value is

$$X_0 = a_0 = \frac{1}{T} \int_0^T x(t) dt = \frac{1}{2\pi} \left(\int_0^\pi A \sin(t) dt + 0 \right) = \frac{A}{2\pi} [-\cos(t)]_0^\pi = \frac{A}{\pi}$$

The exponential Fourier series for the given waveform is

$$\begin{aligned} x(t) &= \sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t} = \sum_{n=-\infty}^{\infty} X_n e^{jnt} \\ &= \dots - \frac{A}{15\pi} e^{-j4t} - \frac{A}{3\pi} e^{-j2t} + j \frac{A}{4} e^{-jt} + \frac{A}{\pi} - j \frac{A}{4} e^{jt} - \frac{A}{3\pi} e^{j2t} - \frac{A}{15\pi} e^{j4t} - \dots \end{aligned}$$

The harmonic amplitudes are [Eq. (3.38)]

$$c_n = \begin{cases} \frac{A}{\pi}, & n = 0 \\ \frac{A}{2}, & n = 1 \\ \frac{2A}{\pi(n^2 - 1)}, & n = 2, 4, 6, \dots \\ 0, & n = 3, 5, 7, \dots \end{cases}$$

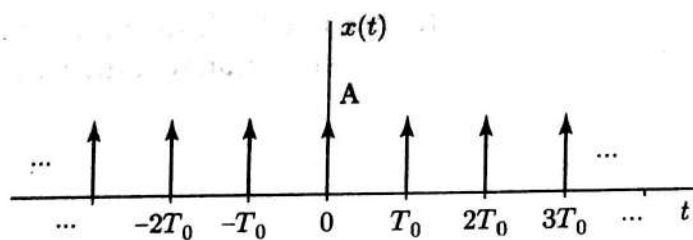
The line spectrum is exactly as plotted in Fig. 3.14.

Example 3.11 Find the exponential Fourier series and sketch the corresponding spectra for the impulse train $x(t) = \delta_{T_0}(t)$ depicted in Fig. 3.15. From this result sketch the line spectrum and write the trigonometric Fourier series for $\delta_{T_0}(t)$.

Solution

The waveform is periodic with period T_0 and fundamental frequency $\omega_0 = 2\pi/T_0$.

$$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_0) \quad (3.41)$$

**Fig. 3.15** Impulse train

By choosing the interval of integration $(-T_0/2, T_0/2)$, the given waveform for one period may be written as

$$x(t) = \delta(t), \quad -\frac{T_0}{2} < t < \frac{T_0}{2}$$

From Eq. (3.27), the exponential Fourier series coefficient is

$$\begin{aligned} X_n &= \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(t) e^{-jn\omega_0 t} dt \\ &= \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} \delta(t) e^{-jn\omega_0 t} dt \end{aligned}$$

From the sampling property (Eq. 1.13), the integral on the RHS is the value of $e^{-jn\omega_0 t}$ at $t = 0$ (where the impulse is located). Therefore,

$$X_n = \frac{1}{T_0}$$

Substitution of this value in Eq. (3.26) yields the desired exponential Fourier series:

$$x(t) = \frac{1}{T_0} \sum_{n=-\infty}^{\infty} e^{jn\omega_0 t} \quad (3.42)$$

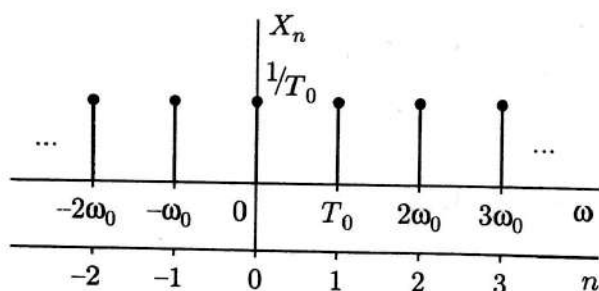
**Fig. 3.16** Exponential Fourier spectra

Figure 3.16 shows the exponential spectrum. To sketch the line spectrum, we use Eq. (3.38) to obtain

$$c_0 = X_0 = \frac{1}{T_0}$$

$$c_n = 2|X_n| = \frac{2}{T_0}, \quad n \geq 1$$

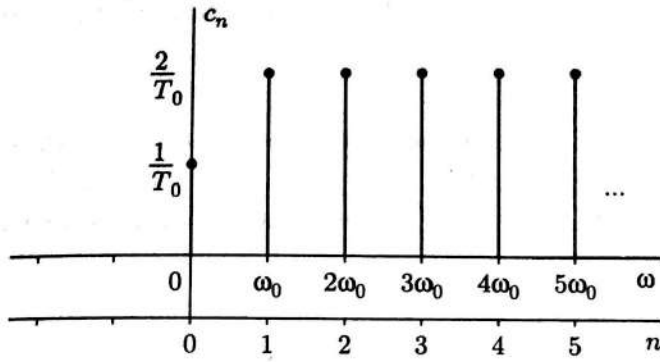


Fig. 3.17 Line spectrum

Figure 3.17 shows the line spectrum. Using Eqs (3.34), (3.35), and (3.36), we may convert an exponential Fourier series into a trigonometric Fourier series:

$$a_0 = X_0 = \frac{1}{T_0}$$

$$a_n = 2\operatorname{Re}\{X_n\} = \frac{2}{T_0}$$

$$b_n = 0$$

3.3 DIRICHLET CONDITIONS

Dirichlet showed that if $x(t)$ satisfies certain conditions (*Dirichlet conditions*), its Fourier series is guaranteed to converge pointwise at all points where $x(t)$ is continuous. These conditions are as follows:

1. Over any period, $x(t)$ must be *absolutely integrable*, i.e.,

$$\int_0^T |x(t)| dt < \infty \quad (3.43)$$

This guarantees that each coefficient X_n will be finite since

$$\begin{aligned} |X_n| &= \frac{1}{T} \left| \int_0^T x(t) e^{-jn\omega_0 t} dt \right| \\ &\leq \frac{1}{T} \int_0^T |x(t) e^{-jn\omega_0 t}| dt = \frac{1}{T} \int_0^T |x(t)| dt \end{aligned}$$

So if

$$\int_0^T |x(t)| dt < \infty$$

then

$$|X_n| < \infty$$

2. In any finite interval of time, $x(t)$ is of bounded variation, i.e., there are no more than a finite number of maxima and minima during any single period of the signal.

3. In any finite interval of time, there are only a finite number of discontinuities. Furthermore, each of these discontinuities is finite.

Thus if a signal $x(t)$ satisfies the Dirichlet conditions, then the corresponding Fourier series is convergent and its sum is $x(t)$, except at any point t_0 at which $x(t)$ is discontinuous. At the point of discontinuity, the sum of the series is the average of the left- and right-hand limits of $x(t)$ at t_0 , i.e.,

$$x(t_0) = \frac{1}{2}[x(t_0^+) + x(t_0^-)]$$

3.4 PROPERTIES OF CONTINUOUS-TIME FOURIER SERIES

Fourier series representations possesses a number of important properties that are useful for developing conceptual insights into such representations, and they can also help to reduce the complexity of the evaluation of the Fourier series of many signals.

We will use a shorthand notation to indicate the relationship between a periodic signal and its Fourier series coefficients. Specifically, suppose that $x(t)$ is periodic signal with period T and fundamental frequency $\omega_0 = 2\pi/T$. Then if the Fourier series coefficients of $x(t)$ are denoted by X_n , we will use the notation

$$x(t) \longleftrightarrow X_n$$

to signify the pairing of a periodic signal with its Fourier series coefficients.

3.4.1 Linearity

If $x(t)$ and $y(t)$ denote two periodic signals with period T and

$$x(t) \longleftrightarrow X_n$$

$$y(t) \longleftrightarrow Y_n$$

then

$$z(t) = ax(t) + by(t) \longleftrightarrow Z_n = aX_n + bY_n \quad (3.44)$$

Proof The exponential Fourier series coefficient of $z(t)$ is

$$\begin{aligned} Z_n &= \frac{1}{T} \int_0^T z(t) e^{-jn\omega_0 t} dt \\ &= \frac{1}{T} \int_0^T [ax(t) + by(t)] e^{-jn\omega_0 t} dt \\ &= a \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt + b \frac{1}{T} \int_0^T y(t) e^{-jn\omega_0 t} dt \\ &= aX_n + bY_n \end{aligned}$$

3.4.2 Time Shifting

When a time shift is applied to a periodic signal $x(t)$, the period T of the signal is preserved. If

$$x(t) \longleftrightarrow X_n$$

then

$$y(t) = x(t - t_0) \longleftrightarrow Y_n = X_n e^{-jn\omega_0 t_0} \quad (3.45)$$

One consequence of this property is that when a signal is shifted in time the *magnitudes* of its Fourier series coefficients remain unaltered, i.e., $|Y_n| = |X_n|$.

Proof By definition

$$Y_n = \frac{1}{T} \int_0^T y(t) e^{-jn\omega_0 t} dt = \frac{1}{T} \int_0^T x(t - t_0) e^{-jn\omega_0 t} dt$$

A change of variables is performed by letting $\tau = t - t_0$, which also yields $d\tau = dt$, $\tau \rightarrow -t_0$ as $t \rightarrow 0$, and $\tau \rightarrow T - t_0$ as $t \rightarrow T$. Therefore,

$$\begin{aligned} Y_n &= \frac{1}{T} \int_{-t_0}^{T-t_0} x(\tau) e^{-jn\omega_0(\tau+t_0)} d\tau \\ Y_n &= \frac{1}{T} \int_{-t_0}^{T-t_0} \underbrace{x(\tau) e^{-jn\omega_0 \tau}}_{X_n} d\tau e^{-jn\omega_0 t_0} = X_n e^{-jn\omega_0 t_0} \end{aligned}$$

$$|Y_n| = |X_n| \quad \blacksquare$$

3.4.3 Frequency Shifting

If

$$x(t) \longleftrightarrow X_n$$

then

$$y(t) = e^{jm\omega_0 t} x(t) \longleftrightarrow Y_n = X_{n-m} \quad (3.46)$$

Proof By definition

$$Y_n = \frac{1}{T} \int_0^T y(t) e^{-jn\omega_0 t} dt = \frac{1}{T} \int_0^T e^{jm\omega_0 t} x(t) e^{-jn\omega_0 t} dt = \frac{1}{T} \int_0^T x(t) e^{-j(n-m)\omega_0 t} dt = X_{n-m}$$

Hence, a frequency shift corresponds to multiplication in time domain by a complex sinusoid whose frequency is equal to the time shift. $\blacksquare$

3.4.4 Time Reversal

If

$$x(t) \longleftrightarrow X_n$$

then

$$y(t) = x(-t) \longleftrightarrow Y_n = X_{-n} \quad (3.47)$$

In other words time reversal applied to a continuous-time signal results in a time reversal of the corresponding sequence of Fourier series coefficients.

Proof By definition

$$Y_n = \frac{1}{T} \int_0^T y(t) e^{-jn\omega_0 t} dt = \frac{1}{T} \int_0^T x(-t) e^{-jn\omega_0 t} dt = \frac{1}{T} \int_0^T x(t) e^{-j(-n)\omega_0 t} dt = X_{-n}$$

An interesting consequence of the time-reversal property is that if $x(t)$ is even then its Fourier series coefficients are also even, i.e.,

$$\text{if } x(-t) = x(t), \text{ then } X_{-n} = X_n \quad (3.48)$$

Similarly, if $x(t)$ is odd, then so are its Fourier series coefficients, i.e.,

$$\text{if } x(-t) = -x(t), \text{ then } X_{-n} = -X_n \quad (3.49)$$

3.4.5 Time Scaling

If

$$x(t) \longleftrightarrow X_n$$

then

$$y(t) = x(at) \longleftrightarrow Y_n = X_n \quad (3.50)$$

Proof Time scaling is an operation that in general changes the period of the underlying signal. Specifically, if $x(t)$ is periodic with period T and fundamental frequency $\omega_0 = 2\pi/T$, then $y(t) = x(at)$, where a is a positive real number, is periodic with period T/a and fundamental frequency $a\omega_0$.

$$\begin{aligned} Y_n &= \frac{1}{T/a} \int_0^{T/a} y(t) e^{-jn a \omega_0 t} dt \\ &= \frac{1}{T/a} \int_0^{T/a} x(at) e^{-jn a \omega_0 t} dt \end{aligned}$$

A change of variables is performed by letting $\tau = at$, which also yields $d\tau = a dt$, $\tau \rightarrow 0$ as $t \rightarrow 0$, and $\tau \rightarrow T$ as $t \rightarrow T/a$. Therefore,

$$Y_n = \frac{1}{T} \int_0^T x(\tau) e^{-jn\omega_0 \tau} d\tau = X_n$$

We emphasize that while the Fourier series coefficients have not changed, the Fourier series representation has changed because of the change in the fundamental frequency. ■

3.4.6 Periodic Convolution

If

$$x(t) \longleftrightarrow X_n$$

$$y(t) \longleftrightarrow Y_n$$

then

$$z(t) = x(t) \circledast y(t) = \frac{1}{T} \int_0^T x(\tau) y(t - \tau) d\tau \longleftrightarrow Z_n = X_n Y_n \quad (3.51)$$

Proof For periodic signals with the same period, a special form of convolution, known as periodic convolution, is defined by the integral:

$$z(t) = \frac{1}{T} \int_0^T x(\tau) y(t - \tau) d\tau$$

It is easy to show that $z(t)$ is periodic with period T and the periodic convolution is commutative and associative. Thus, we can write $z(t)$ in a Fourier series representation with coefficients:

$$\begin{aligned} Z_n &= \frac{1}{T} \int_0^T z(t) e^{-jn\omega_0 t} dt \\ &= \frac{1}{T} \int_0^T \left(\frac{1}{T} \int_0^T x(\tau) y(t - \tau) d\tau \right) e^{-jn\omega_0 t} dt \\ &= \frac{1}{T} \int_0^T x(\tau) \left(\frac{1}{T} \int_0^T y(t - \tau) e^{-jn\omega_0 t} dt \right) d\tau \end{aligned}$$

From the time-shifting property, i.e., if $y(t) \longleftrightarrow Y_n$, then $y(t - \tau) \longleftrightarrow Y_n e^{-jn\omega_0 \tau}$, we have

$$Z_n = \frac{1}{T} \int_0^T \underbrace{x(\tau) e^{-jn\omega_0 \tau}}_{X_n} d\tau Y_n = X_n Y_n$$

The convolution in time transforms to multiplication of the frequency domain representations. ■

3.4.7 Multiplication

If $x(t)$ and $y(t)$ are periodic signal with the same period T and

$$x(t) \longleftrightarrow X_n$$

$$y(t) \longleftrightarrow Y_n$$

then

$$z(t) = x(t)y(t) \longleftrightarrow Z_n = \sum_{k=-\infty}^{\infty} X_k Y_{n-k} \quad (3.52)$$

Z_n may be interpreted as the discrete-time convolution of the X_n and Y_n :

Proof Consider the signal $z(t)$:

$$z(t) = x(t)y(t) = \sum_{k=-\infty}^{\infty} X_k e^{jk\omega_0 t} \sum_{m=-\infty}^{\infty} Y_m e^{jm\omega_0 t} = \sum_{m=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} X_k Y_m e^{j(k+m)\omega_0 t}$$

A change of variables is performed by letting $n = k + m$, which also yields $m = n - k$, $n \rightarrow -\infty$ as $m \rightarrow -\infty$, and $n \rightarrow \infty$ as $m \rightarrow \infty$. Therefore,

$$z(t) = \sum_{n=-\infty}^{\infty} \left(\sum_{k=-\infty}^{\infty} X_k Y_{n-k} \right) e^{jn\omega_0 t} = \sum_{n=-\infty}^{\infty} Z_n e^{jn\omega_0 t}$$

Thus,

$$Z_n = \sum_{k=-\infty}^{\infty} X_k Y_{n-k}$$

3.4.8 Differentiation

If

$$x(t) \longleftrightarrow X_n$$

then

$$y(t) = \frac{dx(t)}{dt} \longleftrightarrow Y_n = jn\omega_0 X_n \quad (3.53)$$

Proof If $x(t)$ is a periodic signal, then we have the Fourier series representation:

$$x(t) = \sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t}$$

Differentiating both sides of this equation gives

$$y(t) = \frac{dx(t)}{dt} = \sum_{n=-\infty}^{\infty} [X_n jn\omega_0] e^{jn\omega_0 t} = \sum_{n=-\infty}^{\infty} Y_n e^{jn\omega_0 t}$$

and thus we conclude that

$$y(t) = \frac{dx(t)}{dt} \longleftrightarrow Y_n = jn\omega_0 X_n$$

Differentiation forces the time-averaged value of the differentiated signal to be zero; hence, the Fourier series coefficient for $n = 0$ is zero.

3.4.9 Integration

If

$$x(t) \longleftrightarrow X_n$$

and $X_0 = 0$ [so that $\int_{-\infty}^t x(t) dt$ is finite valued and periodic], then

$$y(t) = \int_{-\infty}^t x(t) dt \longleftrightarrow Y_n = \frac{1}{jn\omega_0} X_n \quad (3.54)$$

Proof If $x(t)$ is a periodic signal, then we have the Fourier series representation:

$$x(t) = \sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t}$$

Integrating both sides of this equation gives

$$y(t) = \int_{-\infty}^t x(t) dt = \sum_{n=-\infty}^{\infty} \left(X_n \frac{1}{jn\omega_0} \right) e^{jn\omega_0 t} = \sum_{n=-\infty}^{\infty} Y_n e^{jn\omega_0 t}$$

and thus we conclude that

$$y(t) = \int_{-\infty}^t x(t) dt \longleftrightarrow Y_n = \frac{1}{jn\omega_0} X_n$$

The integration attenuates (de-emphasizes) the magnitude of the high-frequency components of the signal. High-frequency components of the signal are the main contributors to its sharp details, such as those occurring at the points of discontinuity. Hence, integration smoothes the signal, and this is one of the reasons it is sometimes called a *smoothing operation*. ■

3.4.10 Conjugation and Conjugate Symmetry

If

$$x(t) \longleftrightarrow X_n$$

then

$$y(t) = x^*(t) \longleftrightarrow Y_n = X_{-n}^* \quad (3.55)$$

Taking the complex conjugate of a periodic signal $x(t)$ has the effect of complex conjugation and time reversal on the corresponding Fourier series coefficients.

Proof By definition

$$\begin{aligned}
 Y_n &= \frac{1}{T} \int_0^T y(t) e^{-jn\omega_0 t} dt = \frac{1}{T} \int_0^T x^*(t) e^{-jn\omega_0 t} dt \\
 &= \left(\frac{1}{T} \int_0^T x(t) e^{jn\omega_0 t} dt \right)^* = \left(\underbrace{\frac{1}{T} \int_0^T x(t) e^{-j(-n)\omega_0 t} dt}_{X_{-n}} \right)^* \\
 &= (X_{-n})^* = X_{-n}^*
 \end{aligned}$$

Case I If $x(t)$ is real, i.e.,
if

$$x^*(t) = x(t)$$

then

$$X_{-n}^* = X_n$$

$$X_{-n} = X_n^* \quad (3.56)$$

the Fourier series coefficients will be *conjugate symmetric*. Also if $x(t)$ is real and even, then from Eqs (3.48) and (3.56), we have

$$X_{-n} = X_n^* = X_n \quad (3.57)$$

That is, if $x(t)$ is real and even, then so are its Fourier series coefficients.

Case II If $x(t)$ is real and odd, then its Fourier series coefficients are purely imaginary and odd. Using Eqs (3.49) and (3.56), we get

$$X_{-n} = X_n^* = -X_n \quad (3.58)$$

Case III Even and odd decomposition of real signals: If $x(t)$ is real and

$$x(t) \longleftrightarrow X_n$$

then

1.

$$x_e(t) = \mathcal{E}\{x(t)\} \longleftrightarrow \text{Re}\{X_n\} \quad (3.59)$$

The even part of a signal $x(t)$ is defined as

$$x_e(t) = \frac{1}{2}[x(t) + x(-t)]$$

Using the linearity property and Eq. (3.54), we get

$$x_e(t) \longleftrightarrow \frac{1}{2}[X_n + X_{-n}]$$

$$x_e(t) \longleftrightarrow \frac{1}{2}[X_n + X_n^*]$$

$$x_e(t) \longleftrightarrow \frac{1}{2}2\text{Re}\{X_n\}$$

$$x_e(t) \longleftrightarrow \text{Re}\{X_n\}$$

and

2.

www.dahiyastech.blogspot.com

$$x_o(t) = \mathcal{O}\{x(t)\} \longleftrightarrow j \text{Im}\{X_n\} \quad (3.60)$$

The odd part of a signal $x(t)$ is defined as

$$x_o(t) = \frac{1}{2}[x(t) - x(-t)]$$

Using the linearity property and Eq. (3.54), we get

$$x_o(t) \longleftrightarrow \frac{1}{2}[X_n - X_{-n}]$$

$$x_o(t) \longleftrightarrow \frac{1}{2}[X_n - X_n^*]$$

$$x_o(t) \longleftrightarrow \frac{1}{2}2j \text{Im}\{X_n\}$$

$$x_o(t) \longleftrightarrow j \text{Im}\{X_n\}$$

■

3.4.11 Parseval's Theorem for Power Signals

If

$$x(t) \longleftrightarrow X_n$$

then

$$\frac{1}{T} \int_0^T |x(t)|^2 dt = \sum_{n=-\infty}^{\infty} |X_n|^2 = X_0^2 + \sum_{n=1}^{\infty} 2|X_n|^2 \quad (3.61)$$

Parseval's relation states that the total average power in a periodic signal equals the sum of the average powers in all of its harmonic components.

Proof Consider the LHS of Eq. (3.61):

$$\begin{aligned}
 \frac{1}{T} \int_0^T |x(t)|^2 dt &= \frac{1}{T} \int_0^T x(t) x^*(t) dt \\
 &= \frac{1}{T} \int_0^T x(t) \left(\sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t} \right)^* dt \\
 &= \frac{1}{T} \int_0^T x(t) \left(\sum_{n=-\infty}^{\infty} X_n^* e^{-jn\omega_0 t} \right) dt \\
 &= \sum_{n=-\infty}^{\infty} X_n^* \left(\frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt \right) \\
 &= \sum_{n=-\infty}^{\infty} X_n^* X_n \\
 \frac{1}{T} \int_0^T |x(t)|^2 dt &= \sum_{n=-\infty}^{\infty} |X_n|^2 = X_0^2 + \sum_{n=1}^{\infty} 2|X_n|^2
 \end{aligned}$$

The result indicates that the total average power of $x(t)$ is the sum of the average power in each harmonic component. ■

3.5 SYSTEMS WITH PERIODIC INPUTS

The response of an LTI system to a sinusoidal input leads to a characterization of system behaviour that is termed the *frequency response* of the system. This characterization is obtained in terms of the impulse response by using convolution and a complex sinusoidal input signal. Let the impulse response of a system be $h(t)$ and the input be $x(t) = e^{j\omega_0 t}$. Then, the convolution integral gives the output as

$$\begin{aligned}
 y(t) &= h(t) * x(t) \\
 &= \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau \\
 &= \int_{-\infty}^{\infty} h(\tau) e^{j\omega_0(t-\tau)} d\tau \\
 &= \int_{-\infty}^{\infty} h(\tau) e^{-j\omega_0 \tau} d\tau e^{j\omega_0 t} \\
 y(t) &= H(\omega_0) e^{j\omega_0 t}
 \end{aligned}$$

where we define

$$H(\omega_0) = \int_{-\infty}^{\infty} h(\tau) e^{-j\omega_0 \tau} d\tau$$

The output of the system is thus a complex sinusoid of the same frequency as the input multiplied by the complex number $H(\omega_0)$ [Fig. 3.18(a)]. An intuitive interpretation of

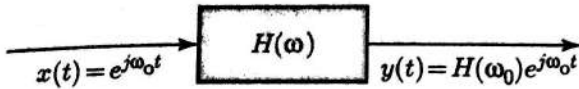


Fig. 3.18(a) Response of an LTI system to complex sinusoid

the sinusoidal steady-state response is obtained by writing the complex-valued frequency response $H(\omega_0)$ in polar form. We have

$$H(\omega_0) = |H(\omega_0)| e^{j\angle H(\omega_0)} \quad (3.62)$$

where $|H(\omega_0)|$ is now termed the *magnitude response* and $\angle H(\omega_0)$ is termed the *phase response* of the system. By substituting this polar form into Eq. (3.62), we may express the output as

$$y(t) = |H(\omega_0)| e^{j(\omega_0 t + \angle H(\omega_0))} \quad (3.63)$$

The system thus modifies the amplitude of the input by $|H(\omega_0)|$ and the phase by $\angle H(\omega_0)$. Knowing $H(\omega_0)$, we can determine whether the system amplifies or attenuates a given sinusoidal component of the input and how much of a phase shift the system adds to that particular component.

We say that the complex sinusoid $x(t) = e^{j\omega_0 t}$ is an *eigenfunction* of the system associated with the *eigenvalue* $H(\omega_0)$. Signals that are eigenfunctions of systems play an important role in LTI system theory. By representing arbitrary signals as weighted superpositions of eigenfunctions, we transform the operation of convolution to multiplication. Thus, if the input to an LTI system is given by

$$x(t) = \sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t} \quad (3.64)$$

then, the output of the system is given by [Fig. 3.18(b)]

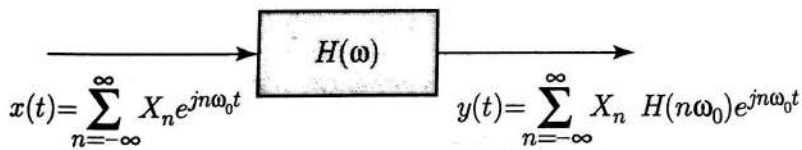


Fig. 3.18(b) Response of an LTI system to a periodic signal

$$y(t) = \sum_{n=-\infty}^{\infty} X_n H(n\omega_0) e^{jn\omega_0 t} = \sum_{n=-\infty}^{\infty} Y_n e^{jn\omega_0 t} \quad (3.65)$$

Therefore, the Fourier series coefficients of the output of an LTI system is given by

$$Y_n = X_n H(n\omega_0) \quad (3.66)$$

SOLVED EXAMPLES

Example 3.12 A continuous-time periodic signal $x(t)$ is real-valued and has a fundamental period $T = 8$. The nonzero Fourier series coefficients for $x(t)$ are

$$X_1 = X_{-1} = 2, \quad X_3 = X_{-3} = 4j$$

Express $x(t)$ in the form

$$x(t) = \sum_{n=-\infty}^{\infty} A_n \cos(\omega_n t + \phi_n)$$

Solution

The signal $x(t)$ is periodic with fundamental frequency $\omega_0 = 2\pi/8 = \pi/4$. Using the Fourier synthesis Eq. (3.26), we have

$$\begin{aligned} x(t) &= \sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t} \\ &= X_1 e^{j\omega_0 t} + X_{-1} e^{-j\omega_0 t} \\ &\quad + X_3 e^{j3\omega_0 t} + X_{-3} e^{-j3\omega_0 t} \\ &= 2e^{j\frac{\pi}{4}t} + 2e^{-j\frac{\pi}{4}t} + 4je^{j3\frac{\pi}{4}t} - 4je^{-j3\frac{\pi}{4}t} \\ &= 4 \left(\frac{e^{j\frac{\pi}{4}t} + e^{-j\frac{\pi}{4}t}}{2} \right) - 8 \left(\frac{e^{j3\frac{\pi}{4}t} - e^{-j3\frac{\pi}{4}t}}{2j} \right) \\ &= 4\cos\left(\frac{\pi}{4}t\right) - 8\sin\left(\frac{3\pi}{4}t\right) \\ &= 4\cos\left(\frac{\pi}{4}t\right) + 8\cos\left(\frac{3\pi}{4}t + \frac{\pi}{2}\right) \end{aligned}$$

Example 3.13 For the continuous-time periodic signal

$$x(t) = 2 + \cos\left(\frac{2\pi}{3}t\right) + 4\sin\left(\frac{5\pi}{3}t\right),$$

determine the fundamental frequency ω_0 and the Fourier series coefficients X_n such that

$$x(t) = \sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t}$$

Solution

Consider the given signal

$$\begin{aligned} x(t) &= 2 + \cos\left(\frac{2\pi}{3}t\right) + 4\sin\left(\frac{5\pi}{3}t\right) \\ &= 2 + \left(\frac{e^{j\frac{2\pi}{3}t} + e^{-j\frac{2\pi}{3}t}}{2} \right) + 4 \left(\frac{e^{j\frac{5\pi}{3}t} - e^{-j\frac{5\pi}{3}t}}{2j} \right) \\ &= 2 + \frac{1}{2}e^{j2\frac{\pi}{3}t} + \frac{1}{2}e^{-j2\frac{\pi}{3}t} + \frac{2}{j}e^{j5\frac{\pi}{3}t} - \frac{2}{j}e^{-j5\frac{\pi}{3}t} \\ x(t) &= 2 + \frac{1}{2}e^{j2\frac{\pi}{3}t} + \frac{1}{2}e^{-j2\frac{\pi}{3}t} - 2je^{j5\frac{\pi}{3}t} + 2je^{-j5\frac{\pi}{3}t} \end{aligned}$$

Comparing with

$$\begin{aligned} x(t) &= X_0 + X_2 e^{j2\omega_0 t} + X_{-2} e^{-j2\omega_0 t} \\ &\quad + X_5 e^{j5\omega_0 t} + X_{-5} e^{-j5\omega_0 t} \end{aligned}$$

The nonzero Fourier series coefficients are

$$X_0 = 2, \quad X_2 = X_{-2} = \frac{1}{2}, \quad X_5 = X_{-5} = -2j$$

Example 3.14 Suppose the periodic signal $x(t)$ has fundamental period T and Fourier coefficients X_n . In a variety of situations, it is easier to calculate the Fourier series coefficients Y_n for $y(t) = dx(t)/dt$ as apposed to calculating X_n directly. Given that

$$\int_T x(t) dt = 2$$

Find an expression for X_n in terms of Y_n and T .

Solution

Given that

$$x(t) \longleftrightarrow X_n$$

Using the differentiation property, Eq. (3.53), we have

$$y(t) = \frac{dx(t)}{dt} \longleftrightarrow Y_n = jn\omega_0 X_n$$

Therefore,

$$X_n = \frac{Y_n}{jn\omega_0}, \quad n \neq 0$$

From Eq. (3.27), we have

$$X_n = \frac{1}{T} \int_T x(t) e^{-jn\omega_0 t} dt$$

for $n = 0$, we have

$$X_n = \frac{1}{T} \int_T x(t) dt = \frac{2}{T}$$

Therefore,

$$X_n = \begin{cases} \frac{2}{T}, & n = 0 \\ \frac{Y_n}{jn\omega_0}, & n \neq 0 \end{cases}$$

Example 3.15 Find the compact trigonometric Fourier series for the periodic signal $x(t)$ shown in Fig. 3.19. Sketch the amplitude and phase spectra for $x(t)$.

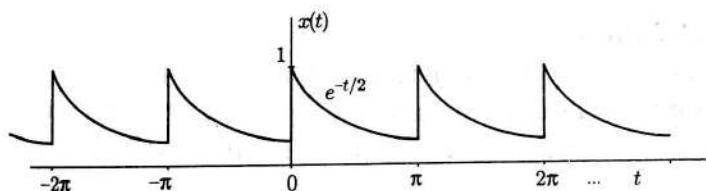


Fig. 3.19 Periodic signal

Solution

The given signal $x(t)$ is periodic with period $T = \pi$ and fundamental frequency $\omega_0 = 2\pi/\pi = 2$ rad/s. The given waveform for one period is defined as

$$x(t) = e^{-t/2}, \quad 0 < t < \pi$$

Since the wave is neither even nor odd, the series will contain both sine and cosine terms.

$$\begin{aligned} a_0 &= \frac{1}{T} \int_0^T x(t) dt = \frac{1}{\pi} \int_0^\pi e^{-t/2} dt \\ &= \frac{1}{\pi} \left[\frac{e^{-t/2}}{-1/2} \right]_0^\pi = \frac{2}{\pi} [1 - e^{-\pi/2}] = 0.504 \end{aligned}$$

Now we determine a_n :

$$\begin{aligned} a_n &= \frac{2}{T} \int_0^T x(t) \cos(n\omega_0 t) dt \\ &= \frac{2}{\pi} \int_0^\pi e^{-t/2} \cos(2nt) dt \\ &= \frac{2}{\pi} I \end{aligned}$$

where

$$\begin{aligned} I &= \int_0^\pi e^{-t/2} \cos(2nt) dt \\ &= \left[\frac{e^{-t/2} \sin(2nt)}{2n} \right]_0^\pi \\ &\quad + \frac{1}{4n} \int_0^\pi e^{-t/2} \sin(2nt) dt \\ &= 0 + \frac{1}{4n} \left[-\frac{e^{-t/2} \cos(2nt)}{2n} \right]_0^\pi \\ &\quad - \frac{1}{16n^2} \int_0^\pi e^{-t/2} \cos(2nt) dt \\ I &= \frac{1}{4n} \left[-\frac{e^{-t/2} \cos(2nt)}{2n} \right]_0^\pi - \frac{1}{16n^2} I \end{aligned}$$

$$I \left(1 + \frac{1}{16n^2} \right) = \frac{1}{8n^2} [1 - e^{-\pi/2}]$$

$$I \left(\frac{16n^2 + 1}{16n^2} \right) = \frac{1}{8n^2} [1 - e^{-\pi/2}]$$

$$I = 2 \frac{[1 - e^{-\pi/2}]}{16n^2 + 1}$$

Substituting I in the expression of a_n gives

$$\begin{aligned} a_n &= 2 \frac{\frac{2}{\pi} [1 - e^{-\pi/2}]}{16n^2 + 1} \\ &= 0.504 \left(\frac{2}{16n^2 + 1} \right) \end{aligned}$$

Now we determine b_n :

$$\begin{aligned} b_n &= \frac{2}{T} \int_0^T x(t) \sin(n\omega_0 t) dt \\ &= \frac{2}{\pi} \int_0^\pi e^{-t/2} \sin(2nt) dt \\ &= 0.504 \left(\frac{8n}{16n^2 + 1} \right) \end{aligned}$$

Substituting the values of a_0 , a_n , and b_n into Eq (3.6), gives

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

$$x(t) = 0.504 \left[1 + \sum_{n=1}^{\infty} \frac{2}{1 + 16n^2} \left(\cos(2nt) + 4n \sin(2nt) \right) \right]$$

Also from Eqs (3.10) and (3.11)

$$c_0 = a_0 = 0.504$$

$$c_n = \sqrt{a_n^2 + b_n^2}$$

$$= 0.504 \sqrt{\frac{4}{(1 + 16n^2)^2} + \frac{64n^2}{(1 + 16n^2)^2}}$$

$$= 0.504 \left(\frac{2}{\sqrt{1 + 16n^2}} \right)$$

$$\theta_n = \tan^{-1} \left(\frac{-b_n}{a_n} \right) = \tan^{-1}(4n) = -\tan^{-1}(4n)$$

Amplitude and phases of the dc and the first six harmonics are computed from the above equations and displayed in Table 3.2. The amplitude and phase spectra are shown in Figs 3.20(a) and (b), respectively.

Table 3.2

n	c_n	θ_n
0	0.504	0
1	0.244	-75.96
2	0.125	-82.87
3	0.084	-85.24
4	0.063	-86.42
5	0.0504	-87.14
6	0.042	-87.61

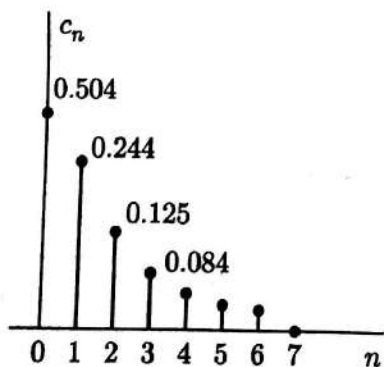


Fig. 3.20(a) Amplitude spectra

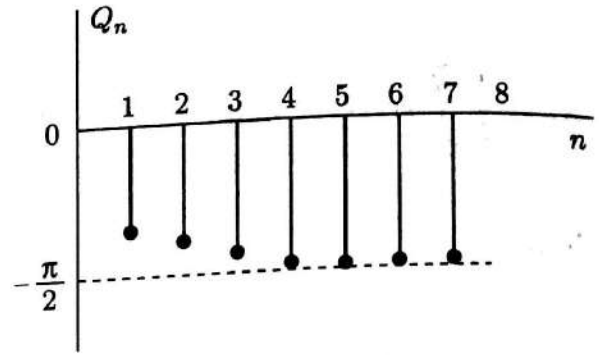


Fig. 3.20(b) Phase spectra

The compact or polar form representation of Fourier series is

$$x(t) = 0.504$$

$$+ 0.504 \sum_{n=1}^{\infty} \frac{2}{\sqrt{1 + 16n^2}} \cos(2nt - \tan^{-1}(4n))$$

Example 3.16 Find the exponential Fourier series for the signal in Fig. 3.19.

Solution

The waveform is periodic with period $T = \pi$ and fundamental frequency $\omega_0 = 2\pi/\pi = 2$ rad/s. The given waveform for one period is defined as

$$x(t) = e^{-t/2}, \quad 0 < t < \pi$$

From Eq. (3.27), the exponential Fourier series coefficient is

$$X_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt = \frac{1}{\pi} \int_0^{\pi} e^{-\frac{t}{2}} e^{-j2nt} dt$$

$$X_n = \frac{1}{\pi} \int_0^{\pi} e^{-\left(\frac{1}{2} + j2n\right)t} dt$$

$$= -\frac{1}{\pi\left(\frac{1}{2} + j2n\right)} \left[e^{-\left(\frac{1}{2} + j2n\right)t} \right]_0^{\pi}$$

$$X_n = \frac{0.504}{1 + j4n}$$

Substituting $n = 0, \pm 1, \pm 2, \dots$ gives

$$X_0 = 0.504$$

$$X_1 = \frac{0.504}{1 + j4} = 0.122 e^{-j75.96}$$

$$|X_1| = 0.122, \angle X_1 = -75.96$$

$$X_{-1} = \frac{0.504}{1 - j4} = 0.122 e^{j75.96}$$

$$|X_{-1}| = 0.122, \angle X_{-1} = 75.96$$

$$X_2 = \frac{0.504}{1 + j8} = 0.0625 e^{-j82.87}$$

$$|X_2| = 0.0625, \angle X_2 = -82.87$$

$$X_{-2} = \frac{0.504}{1 - j8} = 0.0625 e^{j82.87}$$

$$|X_{-2}| = 0.0625, \angle X_{-2} = 82.87$$

Note that X_n and X_{-n} are conjugates, as expected [Eq. (3.55)]. Figure 3.21(a) shows the magnitude spectrum and Fig. 3.21(b) shows the phase spectrum.

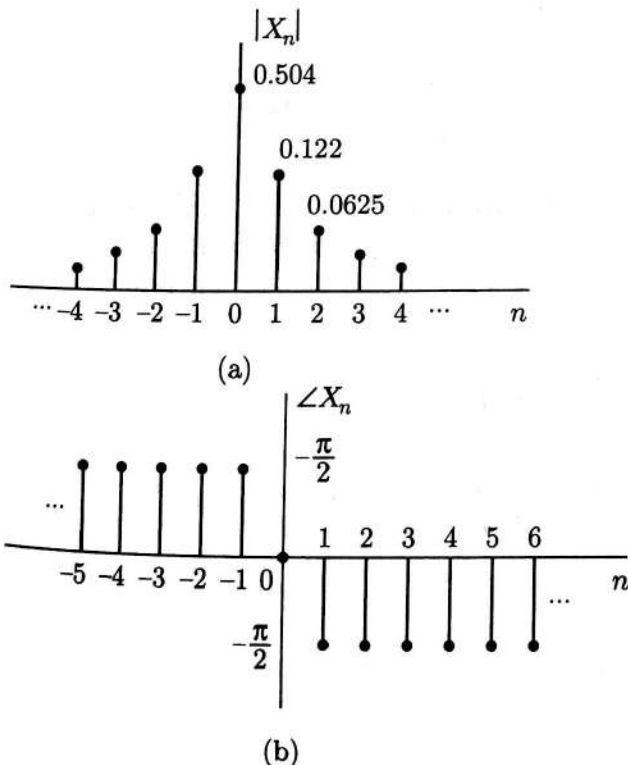


Fig. 3.21 (a) Magnitude spectrum, (b) Phase spectrum

The exponential Fourier series for the given waveform is

$$x(t) = \sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t} = 0.504 \sum_{n=-\infty}^{\infty} \frac{1}{1 + j4n} e^{j2\pi n t}$$

Example 3.17 Suppose we are given the following information about a signal $x(t)$:

1. $x(t)$ is real and odd.
2. $x(t)$ is periodic with period $T = 2$ and has Fourier coefficients X_n .
3. $X_n = 0$ for $|n| > 1$.
4. $\frac{1}{2} \int_0^2 |x(t)|^2 dt = 1$.

Specify two different signals that satisfy these conditions.

Solution

Since $x(t)$ is real and odd (clue 1), its Fourier series coefficients are purely imaginary and odd [Eq. (3.58)]. Therefore,

$$X_n = -X_{-n} \quad \text{and} \quad X_0 = 0$$

Also, since it is given that $X_n = 0$ for $|n| > 1$, the only unknown Fourier series coefficients are X_1 and X_{-1} . Using Parseval's relation,

$$\frac{1}{T} \int_0^T |x(t)|^2 dt = \sum_{n=-\infty}^{\infty} |X_n|^2$$

for the given signal we have

$$\frac{1}{2} \int_0^2 |x(t)|^2 dt = \sum_{n=-1}^1 |X_n|^2 = 1$$

$$|x_1|^2 + |x_{-1}|^2 = 1$$

$$2|X_1|^2 = 1$$

$$|X_1|^2 = \frac{1}{2}$$

Therefore,

$$X_1 = -X_{-1} = j\frac{1}{\sqrt{2}} \quad \text{or} \quad X_1 = -X_{-1} = -j\frac{1}{\sqrt{2}}$$

The two possible signals which satisfy the given information are

$$\begin{aligned}x_1(t) &= j \frac{1}{\sqrt{2}} e^{j\pi t} - j \frac{1}{\sqrt{2}} e^{-j\pi t} \\&= j \frac{1}{\sqrt{2}} 2j \left(\frac{e^{j\pi t} - e^{-j\pi t}}{2j} \right) \\&= -\sqrt{2} \sin(\pi t)\end{aligned}$$

and

$$\begin{aligned}x_2(t) &= -j \frac{1}{\sqrt{2}} e^{j\pi t} + j \frac{1}{\sqrt{2}} e^{-j\pi t} \\&= -j \frac{1}{\sqrt{2}} 2j \left(\frac{e^{j\pi t} - e^{-j\pi t}}{2j} \right) \\&= \sqrt{2} \sin(\pi t)\end{aligned}$$

Example 3.18 Let $x(t)$ be a periodic signal whose Fourier series coefficients are

$$X_n = \begin{cases} 2, & n = 0 \\ j \left(\frac{1}{2} \right)^{|n|}, & \text{otherwise} \end{cases}$$

Use Fourier series properties to answer the following questions:

- Is $x(t)$ real?
- Is $x(t)$ even?
- Is $dx(t)/dt$ even?

Solution

(a) If $x(t)$ is real, then $X_n = X_n^*$. Since $X_n \neq X_n^*$, $x(t)$ is not real.

(b) If $x(t)$ is even, then $X_n = X_{-n}$.

$$X_n = X_{-n} = \begin{cases} 2, & n = 0 \\ j \left(\frac{1}{2} \right)^{|k|}, & \text{otherwise} \end{cases}$$

Since $X_n = X_{-n}$, $x(t)$ is even.

(c) We have

$$y(t) = \frac{dx(t)}{dt} \longleftrightarrow Y_n = jn\omega_0 X_n$$

Therefore,

$$Y_n = \begin{cases} 0, & n = 0 \\ -n \left(\frac{1}{2} \right)^{|n|} \omega_0, & \text{otherwise} \end{cases}$$

and

$$Y_{-n} = \begin{cases} 0, & n = 0 \\ n \left(\frac{1}{2} \right)^{|n|} \omega_0, & \text{otherwise} \end{cases}$$

Since $Y_n \neq Y_{-n}$, $y(t) = dx(t)/dt$ is not even.

Example 3.19 Let $x(t)$ be a periodic signal with fundamental period T and Fourier series coefficients X_n . Derive the Fourier series coefficients of each of the following signals in terms of X_n :

- $x(t - t_0) + x(t + t_0)$
- $\mathcal{E}\{x(t)\} = x_e(t)$
- $\text{Re}\{x(t)\}$
- $\frac{d^2 x(t)}{dt^2}$
- $x(3t - 1)$

Solution

The Fourier series coefficients of $x(t)$ is X_n , i.e.,

$$x(t) \longleftrightarrow X_n$$

(a) $x(t - t_0)$ is also periodic with period T and fundamental frequency $\omega_0 = 2\pi/T$. The Fourier series coefficients A_n of $x(t - t_0)$ are

$$A_n = \frac{1}{T} \int_0^T x(t - t_0) e^{-jn\omega_0 t} dt$$

A change of variables is performed by letting $\tau = (t - t_0)$, which also yields $d\tau = dt$, $\tau \rightarrow -t_0$ as $t \rightarrow 0$, and $\tau \rightarrow (T - t_0)$ as $t \rightarrow T$. Therefore,

$$A_n = \frac{1}{T} \int_{-t_0}^{T-t_0} x(\tau) e^{-jn\omega_0(\tau+t_0)} d\tau$$

$$A_n = \frac{1}{T} \int_{-t_0}^{T-t_0} x(\tau) e^{-jn\omega_0 \tau} d\tau e^{-jn\omega_0 t_0}$$

$$A_n = X_n e^{-jn\omega_0 t_0}$$

Similarly, using the time-shifting property, the Fourier series coefficients B_n of $x(t + t_0)$ are

$$B_n = X_n e^{jn\omega_0 t_0}$$

Finally, the Fourier series coefficients Y_n of $x(t - t_0) + x(t + t_0)$ are

$$\begin{aligned} Y_n &= A_n + B_n = X_n e^{-jn\omega_0 t_0} + X_n e^{jn\omega_0 t_0} \\ &= 2X_n \left(\frac{e^{-jn\omega_0 t_0} + e^{jn\omega_0 t_0}}{2} \right) \\ &= 2\cos(n\omega_0 t_0) X_n \end{aligned}$$

(b) Note that

$$\mathcal{E}\{x(t)\} = x_e(t) = \frac{x(t) + x(-t)}{2}$$

The Fourier series coefficients of $x(-t)$ are

$$\begin{aligned} A_n &= \frac{1}{T} \int_0^T x(-t) e^{-jn\omega_0 t} dt \\ &= \frac{1}{T} \int_0^T x(t) e^{-j(-n)\omega_0 t} dt \\ &= X_{-n} \end{aligned}$$

Therefore, the Fourier series coefficients Y_n of $x_e(t)$ are

$$Y_n = \frac{X_n + X_{-n}}{2}$$

(c) Note that

$$\text{Re}\{x(t)\} = \frac{x(t) + x^*(t)}{2}$$

The Fourier series coefficients A_n of $x^*(t)$ are

$$\begin{aligned} A_n &= \frac{1}{T} \int_0^T x^*(t) e^{-jn\omega_0 t} dt \\ &= \left(\frac{1}{T} \int_0^T x(t) e^{jn\omega_0 t} dt \right)^* \\ &= \left(\frac{1}{T} \int_0^T x(t) e^{-j(-n)\omega_0 t} dt \right)^* \\ A_n &= (X_{-n})^* = X_{-n}^* \end{aligned}$$

Therefore, the Fourier series coefficients of $\text{Re}\{x(t)\}$ are

$$\begin{aligned} Y_n &= \frac{X_n + A_n}{2} \\ &= \frac{X_n + X_{-n}^*}{2} \end{aligned}$$

(d) The Fourier series synthesis equation gives

$$x(t) = \sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t}$$

Differentiating both sides w.r.t. t twice gives

$$\frac{d^2 x(t)}{dt^2} = \sum_{n=-\infty}^{\infty} (-n^2 \omega_0^2 X_n) e^{jn\omega_0 t}$$

By inspection, we know that the Fourier series coefficients of $d^2 x(t)/dt^2$ are $-n^2 \omega_0^2 X_n$.

(e) The period of $x(3t - 1)$ is $T/3$ and fundamental frequency $3\omega_0$. The Fourier series coefficients Y_n of $y(t) = x(3t - 1)$ are

$$\begin{aligned} Y_n &= \frac{1}{T/3} \int_0^{T/3} y(t) e^{-jn3\omega_0 t} dt \\ &= \frac{1}{T/3} \int_0^{T/3} x(3t - 1) e^{-jn3\omega_0 t} dt \end{aligned}$$

A change of variables is performed by letting $\tau = (3t - 1)$, which also yields $dt = 1/3 d\tau$, $\tau \rightarrow -1$ as $t \rightarrow 0$, and $\tau \rightarrow T - 1$ as $t \rightarrow T/3$. Therefore,

$$\begin{aligned} Y_n &= \frac{1}{T} \int_{-1}^{T-1} x(\tau) e^{-jn3\omega_0 \frac{(\tau+1)}{3}} d\tau \\ &= \frac{1}{T} \int_{-1}^{T-1} x(\tau) e^{-jn\omega_0 \tau} d\tau e^{-jn\omega_0} \\ &= X_n e^{-jn\omega_0} \end{aligned}$$

Example 3.20 Suppose we are given the following information about a continuous-time periodic signal with period 3 and Fourier coefficients X_n :

1. $X_n = X_{n+2}$.
2. $X_n = X_{-n}$.
3. $\int_{-0.5}^{0.5} x(t) dt = 1$.
4. $\int_{0.5}^{1.5} x(t) dt = 2$.

Determine $x(t)$.

Solution

The continuous-time signal $x(t)$ is periodic with period $T = 3$ and fundamental frequency $\omega_0 = 2\pi/T = 2\pi/3$. Since $X_n = X_{-n}$, we require that $x(t) = x(-t)$,

i.e., $x(t)$ must be an even signal. Also, note that

$$X_n = X_{n+2}$$

$$\frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt = \frac{1}{T} \int_0^T x(t) e^{-j(n+2)\omega_0 t} dt$$

$$\frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt = \frac{1}{T} \int_0^T [x(t) e^{-j2\omega_0 t}] e^{-jn\omega_0 t} dt$$

we require that

$$x(t) = x(t) e^{-j2\omega_0 t}$$

or

$$e^{-j2\omega_0 t} = 1$$

$$e^{-j2\omega_0 t} = 1 = e^{j2m\pi}, \quad m = 0, \pm 1, \pm 2, \dots$$

The above equation indicates that

$$e^{-j2\omega_0 t} = \begin{cases} 1, & t = 0, \pm \frac{3}{2}, 3, \pm \frac{9}{2}, \dots \\ 0, & \text{otherwise} \end{cases}$$

Since $\int_{-0.5}^{0.5} x(t) dt = 1$, we may conclude that $x(t) = \delta(t)$ for $-0.5 \leq t \leq 0.5$. Also since $\int_{0.5}^{1.5} x(t) dt = 2$, we may conclude that $x(t) = 2\delta(t - 3/2)$ in the range $0.5 \leq t \leq 1.5$. Therefore, $x(t)$ may be written as

$$x(t) = \sum_{k=-\infty}^{\infty} \delta(t - 3k) + 2 \sum_{k=-\infty}^{\infty} \delta(t - 3k - 3/2)$$

Example 3.21 Show that the Fourier series coefficients of the signal

$$z(t) = x(t)y(t) = \sum_{n=-\infty}^{\infty} Z_n e^{jn\omega_0 t}$$

are given by the discrete convolution

$$Z_n = \sum_{k=-\infty}^{\infty} X_k Y_{n-k}$$

Solution

$$\begin{aligned} z(t) &= x(t)y(t) \\ &= \sum_{k=-\infty}^{\infty} X_k e^{jk\omega_0 t} \sum_{n=-\infty}^{\infty} Y_n e^{jn\omega_0 t} \\ &= \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} X_k Y_n e^{j(k+n)\omega_0 t} \end{aligned}$$

A change of variables is performed by letting $m = (k + n)$, which also yields $n = (m - k)$, $m \rightarrow -\infty$ as $n \rightarrow -\infty$, and $m \rightarrow \infty$ as $n \rightarrow \infty$. Therefore,

$$\begin{aligned} z(t) &= \sum_{m=-\infty}^{\infty} \left(\sum_{k=-\infty}^{\infty} X_k Y_{m-k} \right) e^{jm\omega_0 t} \\ &= \sum_{m=-\infty}^{\infty} Z_m e^{jm\omega_0 t} \end{aligned}$$

Thus,

$$Z_n = \sum_{k=-\infty}^{\infty} X_k Y_{n-k}$$

Example 3.22 Let $x(t)$ be a real-valued signal with fundamental period T and Fourier series coefficient X_n .

- Show that $X_n = X_{-n}^*$ and X_0 must be real.
- Show that if $x(t)$ is even, then its Fourier coefficients must be real and even.
- Show that if $x(t)$ is odd, then its Fourier series coefficients are imaginary and odd and $X_0 = 0$.
- Show that the Fourier series coefficients of the even part of $x(t)$ are equal to

$$\operatorname{Re}\{x(t)\}$$

- Show that the Fourier series coefficients of the odd part of $x(t)$ are equal to

$$j\operatorname{Im}\{x(t)\}$$

Solution

- From conjugation and conjugate symmetry property, i.e., Eq. (3.55), we know that Fourier series coefficients of $x^*(t)$ are X_{-n}^* , and we also know that if $x(t)$ is real, then $x(t) = x^*(t)$. Therefore $X_n = X_{-n}^*$.

This implies $X_0 = X_0^*$. Therefore, X_0 must be real.
 (b) Let $x(t)$ be a real and even signal:

$$x(t) = x(-t)$$

From Eq. (3.47), the Fourier series coefficients of $x(-t)$ are X_{-n} . Therefore,

$$X_n = X_n^* = X_{-n}$$

(c) Let $x(t)$ be a real and odd signal:

$$x(t) = -x(-t)$$

From Eq. (3.47), the Fourier series coefficients of $-x(-t)$ are $-X_{-n}$. Therefore,

$$X_n = X_n^* = -X_{-n}$$

Note that for $n = 0$, the above equation requires that $X_0 = -X_0$, we may conclude that $X_0 = 0$.

(d) See the proof of Eq. (3.59).

(e) See the proof of Eq. (3.60).

Example 3.23 Suppose we are given the following information about a signal $x(t)$:

1. $x(t)$ is real.
2. $x(t)$ is periodic with period $T = 6$ and has Fourier coefficients X_n .
3. $X_n = 0$ for $n = 0$ and $n > 2$.
4. $x(t) = -x(t - 3)$.
5. $\frac{1}{6} \int_{-3}^3 |x(t)|^2 dt = \frac{1}{2}$.
6. X_1 is a positive real number.

Show that $x(t) = A \cos(Bt + C)$, and determine the values of the constants A , B , and C .

Solution

The given signal $x(t)$ is periodic with period $T = 6$ and fundamental frequency $\omega_0 = 2\pi/6 = \pi/3$. Since $x(t)$ is real, $X_n = X_n^*$. Since $X_n = 0$ for $n = 0$ and $n > 2$, the only unknown Fourier series coefficients are X_1 , X_{-1} , X_2 , and X_{-2} . Now $x(t)$ is

$$x(t) = X_1 e^{j\omega_0 t} + X_{-1} e^{-j\omega_0 t} + X_2 e^{j2\omega_0 t} + X_{-2} e^{-j2\omega_0 t}$$

Since X_1 is a positive real number, $X_1 = X_{-1}$.

$$\begin{aligned} x(t) &= 2X_1 \left(\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} \right) \\ &\quad + X_2 e^{j2\omega_0 t} + X_{-2} e^{-j2\omega_0 t} \\ x(t) &= 2X_1 \cos(\omega_0 t) + X_2 e^{j2\omega_0 t} + X_{-2} e^{-j2\omega_0 t} \\ x(t) &= 2X_1 \cos(\pi t/3) + X_2 e^{j2\pi t/3} + X_{-2} e^{-j2\pi t/3} \end{aligned}$$

From this, we get

$$\begin{aligned} x(t-3) &= 2X_1 \cos\left(\frac{\pi}{3}(t-3)\right) \\ &\quad + X_2 e^{j2\pi(t-3)/3} + X_{-2} e^{-j2\pi(t-3)/3} \\ x(t-3) &= 2X_1 \cos\left(\frac{\pi}{3}t - \pi\right) \\ &\quad + X_2 e^{j2\pi t/3} e^{-j2\pi} + X_{-2} e^{-j2\pi t/3} e^{j2\pi} \\ x(t-3) &= -2X_1 \cos(\pi t/3) \\ &\quad + X_2 e^{j2\pi t/3} + X_{-2} e^{-j2\pi t/3} \end{aligned}$$

Now, if we need $x(t) = -x(t-3)$, then $X_2 = X_{-2} = 0$, and

$$x(t) = 2X_1 \cos(\pi t/3)$$

Now using Parseval's relation on clue 5, we get

$$\begin{aligned} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt &= \sum_{n=-\infty}^{\infty} |X_n|^2 \\ \frac{1}{6} \int_{-3}^3 |x(t)|^2 dt &= |X_1|^2 + |X_{-1}|^2 = \frac{1}{2} \\ 2|X_1|^2 &= \frac{1}{2} \\ |X_1| &= \frac{1}{2} \end{aligned}$$

Since X_1 is positive, we have $X_1 = X_{-1} = 1/2$. Therefore,

$$x(t) = \cos(\pi t/3)$$

Example 3.24 Consider a continuous-time system whose frequency response is

$$H(j\omega) = \begin{cases} 1, & |\omega| \leq 100 \\ 0, & |\omega| > 100 \end{cases}$$

When the input to this system is a signal $x(t)$ with the fundamental period $T = \pi/6$ and Fourier series coefficient X_n , it is found that

(a) $y(t) = x(t)$. Comment on the Fourier series coefficients of $x(t)$.

(b) $y(t) = 0$. Comment on the Fourier series coefficients of $x(t)$.

Solution

The given signal $x(t)$ is periodic with period $T = \pi/6$, fundamental frequency $\omega_0 = 12$, and $x(t)$ is represented as

$$x(t) = \sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t}$$

From Eqs (3.65) and (3.66), the output of the system is

$$y(t) = \sum_{n=-\infty}^{\infty} X_n H(jn\omega_0) e^{jn\omega_0 t}$$

(a) Since $H(j\omega) = 0$ for $\omega > 100$, the largest value of $|n|$ for which $y(t) = x(t)$ or for which $X_n \neq 0$ should be such that

$$|n|\omega_0 \leq 100$$

$$12|n| \leq 100$$

$$|n| \leq 8$$

Therefore, the Fourier series coefficients $X_n \neq 0$ for $n \leq 8$.

(b) For $y(t) = 0$, the Fourier series coefficients $X_n = 0$ for $n \leq 8$.

Example 3.25 The trigonometric Fourier spectra (line spectrum) of a certain periodic signal $x(t)$ are shown in Fig. 3.22. By inspecting these spectra, sketch the corresponding exponential Fourier spectra and verify your results analytically.

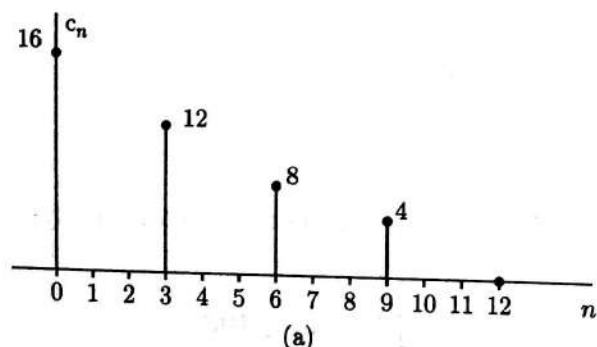


Fig. 3.22(a) Amplitude spectrum

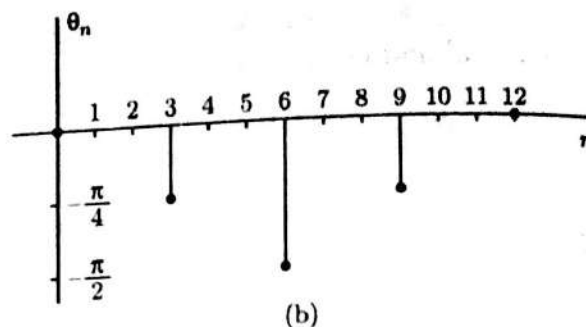


Fig. 3.22(b) Phase spectrum

Solution

The trigonometric spectral components exist at frequencies 0, 3, 6, and 9. The exponential spectral components exist at 0, 3, 6, 9 and $-3, -6, -9$.

First consider the amplitude spectrum. From Eq. (3.38), we have

$$X_0 = c_0 = 16.$$

Now $|X_n|$ is an even function of ω , and from Eq. (3.37), we have

$$|X_n| = |X_{-n}| = c_n/2$$

Thus, all the remaining spectrum $|X_n|$ for positive n is half the trigonometric amplitude spectrum c_n , and the spectrum $|X_n|$ for negative n is a reflection about the vertical axis of the spectrum for positive n , as shown in Fig. 3.23(a).

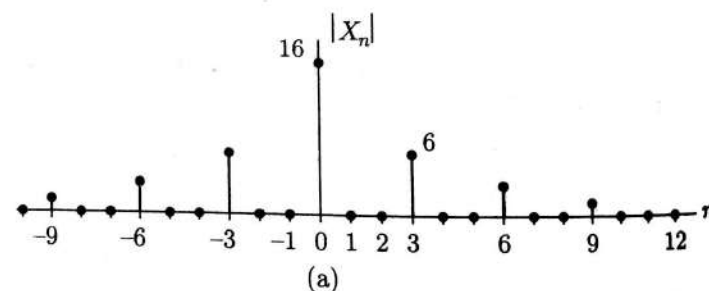


Fig. 3.23(a) Magnitude spectrum

The phase spectrum is an odd function. Therefore, the phase spectrum $\angle X_n = \theta_n$ for positive n and is $-\theta_n$ for negative n as depicted in Fig. 3.23(b).

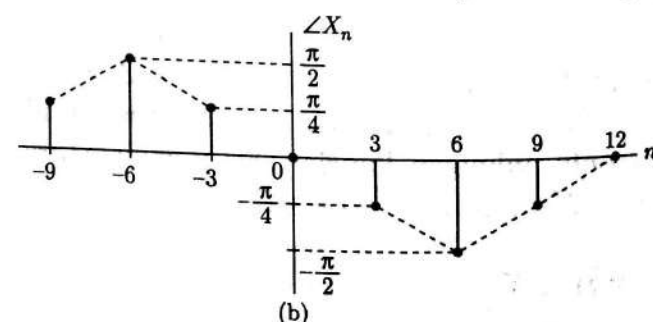


Fig. 3.23(b) Phase spectrum

We shall now verify that both sets of spectra represent the same signal. The amplitude and the phase of the component of frequency 3 are 12 and $-\pi/4$, respectively. Therefore, this component can be expressed as $12 \cos(3t - \pi/4)$.

The amplitude and the phase of the component of frequency 6 are 8 and $-\pi/2$, respectively. Therefore, this component can be expressed as $8 \cos(6t - \pi/2)$.

The amplitude and the phase of the component of frequency 9 are 4 and $-\pi/4$, respectively. Therefore, this component can be expressed as $4 \cos(9t - \pi/4)$. We can write the Fourier series for $x(t)$ as

$$x(t) = 16 + 12 \cos(3t - \pi/4) + 8 \cos(6t - \pi/2) + 4 \cos(9t - \pi/4)$$

Now consider the exponential spectra shown in Fig. 3.23(a) and (b). They contain components of frequencies 0, ± 3 , ± 6 , and ± 9 . The dc component is $X_0 = 16$. The component e^{j3t} (frequency 3) has magnitude 6 and angle $-\pi/4$. Therefore, this component strength is $6e^{-j\pi/4}$, and it can be expressed as $(6e^{-j\pi/4})e^{j3t}$. Similarly, the component of frequency -3 is $(6e^{j\pi/4})e^{-j3t}$. Proceeding in this manner, the signal corresponding to the spectra shown in Fig. 3.23, is

$$\begin{aligned} x(t) &= 16 + [6e^{-j\pi/4} e^{j3t} + 6e^{j\pi/4} e^{-j3t}] \\ &\quad + [4e^{-j\pi/2} e^{j6t} + 4e^{j\pi/2} e^{-j6t}] \\ &\quad + [2e^{-j\pi/4} e^{j9t} + 2e^{j\pi/4} e^{-j9t}] \\ &= 16 + 6[e^{j(3t-\pi/4)} + e^{-j(3t-\pi/4)}] \\ &\quad + 4[e^{j(6t-\pi/2)} + e^{-j(6t-\pi/2)}] \\ &\quad + 2[e^{j(9t-\pi/4)} + e^{-j(9t-\pi/4)}] \\ &= 16 + 12 \cos(3t - \pi/4) + 8 \cos(6t - \pi/2) \\ &\quad + 4 \cos(9t - \pi/4) \end{aligned}$$

Example 3.26 Find the exponential Fourier series and sketch the corresponding Fourier spectrum X_n versus ω for the full-wave rectified sine wave depicted in Fig. 3.24.

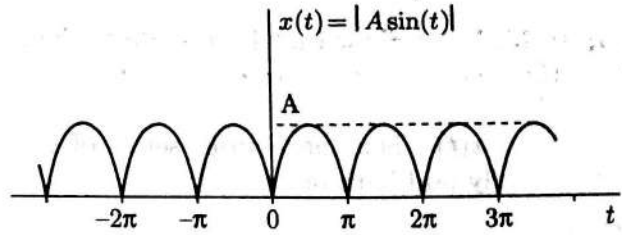


Fig. 3.24 Full-wave rectified sine wave

Solution

The waveform is periodic with period $T = \pi$ and fundamental frequency $\omega_0 = 2\pi/\pi = 2$ rad/s. The given waveform for one period is defined as

$$x(t) = A \sin(t), \quad 0 < t < \pi$$

From Eq. (3.27), the exponential Fourier series coefficient is

$$\begin{aligned} X_n &= \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt \\ &= \frac{1}{\pi} \int_0^\pi A \sin(t) e^{-j2nt} dt \\ &= \frac{A}{\pi} \int_0^\pi e^{-j2nt} \sin(t) dt \\ X_n &= \frac{2A}{\pi(1-4n^2)} \end{aligned}$$

Figure 3.25 shows the magnitude spectrum.

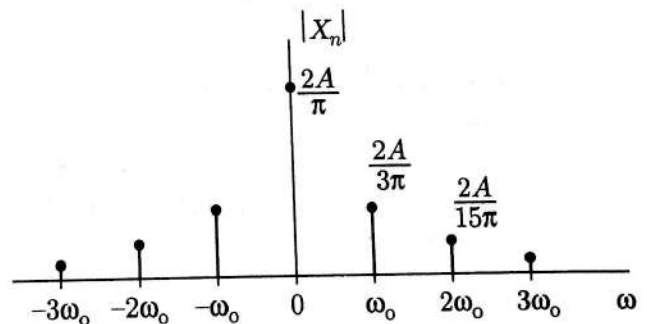


Fig. 3.25 Magnitude spectrum

The exponential Fourier series for the given waveform is

$$x(t) = \sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t} = \frac{2A}{\pi} \sum_{n=-\infty}^{\infty} \frac{1}{1-4n^2} e^{j2nt}$$

Example 3.27 (a) Show that for an odd periodic function, the Fourier series contains only sine terms. (b) Show that if $x(t)$ is periodic with period T , and $x(t \pm T/2) = -x(t)$, then the Fourier series of $x(t)$ will contain only odd harmonics.

Solution

(a) See the proof of Eq. (3.22) in Section 3.2.4.

(b) See the proof of Eq. (3.23) in Section 3.2.4.

Example 3.28 Let $z(t)$ be a real signal and $z(t) = x(t) + y(t)$. Find a condition so that

$$\int_{-\infty}^{\infty} |z(t)|^2 dt = \int_{-\infty}^{\infty} |x(t)|^2 dt + \int_{-\infty}^{\infty} |y(t)|^2 dt$$

or

$$E_z = E_x + E_y$$

Solution

$$E_z = \int_{-\infty}^{\infty} |z(t)|^2 dt$$

$$= \int_{-\infty}^{\infty} |x(t) + y(t)|^2 dt$$

$$= \int_{-\infty}^{\infty} |x(t)|^2 dt + \int_{-\infty}^{\infty} |y(t)|^2 dt$$

$$+ \int_{-\infty}^{\infty} x(t)y^*(t) dt + \int_{-\infty}^{\infty} x^*(t)y(t) dt$$

If $x(t)$ and $y(t)$ are orthogonal, then the two integrals of the products $x(t)y^*(t)$ and $x^*(t)y(t)$ are zero, and

$$\int_{-\infty}^{\infty} |z(t)|^2 dt = \int_{-\infty}^{\infty} |x(t)|^2 dt + \int_{-\infty}^{\infty} |y(t)|^2 dt$$

$$E_z = E_x + E_y$$

The energy of the sum of two orthogonal signals is equal to the sum of the energies of the two signals.

SUMMARY

- Two functions $\phi_i(t)$ and $\phi_j(t)$ are orthogonal over an interval $[0, T]$ if

$$\int_0^T \phi_i(t)\phi_j(t) dt = \begin{cases} E_i, & i = j \\ 0, & i \neq j \end{cases}$$

and are orthonormal over an interval $[0, T]$ if $E_i = 1$ for all i .

- Any arbitrary signal $x(t)$ can be expanded over an interval $[0, T]$ in terms of the orthogonal basis functions $\{\phi_n(t)\}$ as

$$x(t) = \sum_{n=-\infty}^{\infty} X_n \phi_n(t) dt$$

where

$$X_n = \frac{1}{E_n} \int_0^T x(t)\phi_n(t) dt$$

- The complex exponentials

$$\phi_n(t) = e^{jn\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T}$$

are orthogonal over the interval $[0, T]$.

- A periodic signal $x(t)$, of period T , can be expanded in an exponential Fourier series as

$$x(t) = \sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t}$$

where X_n are called Fourier series coefficients and are given by

$$X_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt$$

- The plot of $|X_n|$ versus $n\omega_0$ is called the magnitude spectrum.
- The plot of $\angle X_n$ versus $n\omega_0$ is called the phase spectrum.

Table 3.3 Fourier Series Representation

Series form	Coefficient computation	Conversion formulae
Trigonometric $x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$	$a_0 = \frac{1}{T} \int_0^T x(t) dt$ $a_n = \frac{2}{T} \int_0^T x(t) \cos(n\omega_0 t) dt$ $b_n = \frac{2}{T} \int_0^T x(t) \sin(n\omega_0 t) dt$	$a_0 = c_0 = X_0$ $a_n - jb_n = c_n e^{j\theta_n} = 2X_n$ $a_n + jb_n = c_n e^{-j\theta_n} = 2X_{-n}$
Compact trigonometric (polar form) $x(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t + \theta_n)$	$c_0 = a_0$ $c_n = \sqrt{a_n^2 + b_n^2}$ $\theta_n = \tan^{-1} \left(\frac{-b_n}{a_n} \right)$	$c_0 = X_0$ $c_n = 2 X_n , \quad n \geq 1$ $\theta_n = \angle X_n$
Exponential $x(t) = \sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t}$	$X_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt$	

7. For periodic signals, both the magnitude and phase spectra are line spectra. For real-valued signals, the magnitude spectrum has even symmetry and the phase has odd symmetry.

8. If signal $x(t)$ is real-valued signal, then it can be expanded in trigonometric series of the form

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

where a_0 , a_n , and b_n are the Fourier series coefficients.

9. The relation between the trigonometric-series coefficients and the exponential series coefficients is given by

$$a_0 = X_0$$

$$a_n = 2\operatorname{Re}\{X_n\} \quad \text{or} \quad a_n = X_n + X_{-n}$$

$$b_n = -2\operatorname{Im}\{X_n\} \quad \text{or} \quad b_n = j(X_n - X_{-n})$$

$$X_n = \frac{a_n - jb_n}{2}$$

10. An alternative form (the polar form) of the Fourier series is

$$x(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t + \theta_n)$$

where

$$c_0 = a_0$$

$$c_n = \sqrt{a_n^2 + b_n^2} \quad \text{for } n \geq 1$$

$$\theta_n = \tan^{-1} \frac{-b_n}{a_n}$$

or

$$c_n = \begin{cases} 2|X_n|, & n \geq 1 \\ X_0, & n = 0 \end{cases}$$

$$\angle X_n = \theta_n = \tan^{-1} \frac{-b_n}{a_n}$$

11. For the Fourier series to converge, the signal $x(t)$ must be absolutely integrable, have only a

finite number of maxima and minima, and have a finite number of discontinuities over any period. This set of conditions is known as Dirichlet conditions.

12. The response $y(t)$ of an LTI system to the periodic input $x(t)$ is

$$y(t) = \sum_{n=-\infty}^{\infty} X_n H(n\omega_0) e^{jn\omega_0 t}$$

where $H(jn\omega_0)$ is the frequency response of the system at $\omega = n\omega_0$ and X_n are the Fourier series coefficients of the input $x(t)$.

13. Representing $x(t)$ by a finite series results in an overshoot behaviour at the points of discontinuity. This phenomenon is known as Gibbs's phenomenon.
14. Table 3.3 shows the Fourier series representation of a periodic signal of period T and fundamental frequency $\omega_0 = 2\pi/T$.

MULTIPLE-CHOICE QUESTIONS

- If the Fourier series coefficients of a signal are periodic, then the signal must be
 - continuous time, periodic
 - discrete time, periodic
 - continuous time, nonperiodic
 - discrete time, nonperiodic
- If a periodic function $f(t)$ of period T satisfies $f(t) = -f(t + T/2)$, then in its Fourier series expansion,
 - the constant term will be zero
 - there will be no cosine terms
 - there will be no sine terms
 - there will be no even harmonics
- A periodic signal which can be expanded in Fourier series
 - is a power signal
 - is an energy signal
 - is neither a power signal or an energy signal
 - can be either a power signal or an energy signal depending on the nature
- The Fourier series of a real, even periodic signal will contain only
 - cosine terms
 - sine terms
 - even terms
 - odd harmonics
- Fourier series of any periodic signal $x(t)$ can only be obtained if
 - $\int_0^T |x(t)| dt < \infty$
 - finite number of discontinuities within finite time interval T
 - both (a) and (b)
 - infinite number of discontinuities
- Even signal satisfies
 - $x(t) = x(-t)$
 - $x(t) = -x(-t)$
 - $x(t) = -x(t + T/2)$
 - $x(t) = x(t + T/2)$
- Odd signal satisfies
 - $x(t) = x(-t)$
 - $x(t) = -x(-t)$
 - $x(t) = -x(t + T/2)$
 - $x(t) = x(t + T/2)$
- For half-wave (odd) symmetry, with T_0 is the period of $x(t)$, which one of the following is correct?
 - $x(t \pm T_0/2) = -x(t)$
 - $x(t \pm T_0/2) = x(t)$
 - $x(t \pm T_0) = -x(t)$
 - $x(t \pm T_0) = x(t)$
- The Fourier series representations are based on using
 - constant coefficients
 - only cosine functions
 - only sine functions
 - orthogonal functions
- What is the magnitude of the exponential Fourier series coefficient of the fundamental term of the periodic signal

$$x(t) = 3 + 2 \cos(3t) + \sin(3t)$$
 - $3/2$
 - $\sqrt{5}/2$
 - $1/2$
 - 1

PROBLEMS

3.1 Consider the functions

$$\phi_1(t) = e^{-|t|} \quad \text{and} \quad \phi_2(t) = 1 - A e^{-2|t|}$$

Determine the constant A such that $\phi_1(t)$ and $\phi_2(t)$ are orthogonal over the interval $(-\infty, \infty)$.

3.2 Compute the exponential Fourier series coefficient of

$$x(t) = \sum_{m=-\infty}^{\infty} \left[\delta\left(t - \frac{m}{2}\right) + \delta\left(t - 3\frac{m}{2}\right) \right]$$

3.3 Find the trigonometric Fourier series for periodic signals shown in Figs 3.26(a) and (b). Sketch their amplitude and phase spectra.

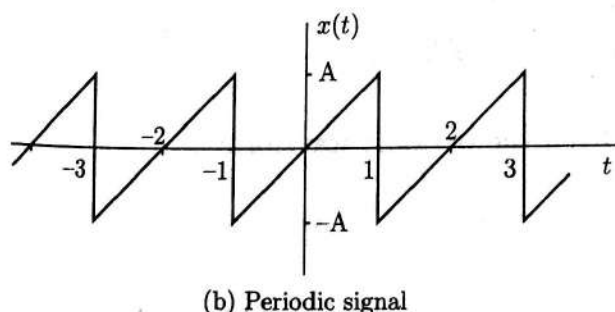
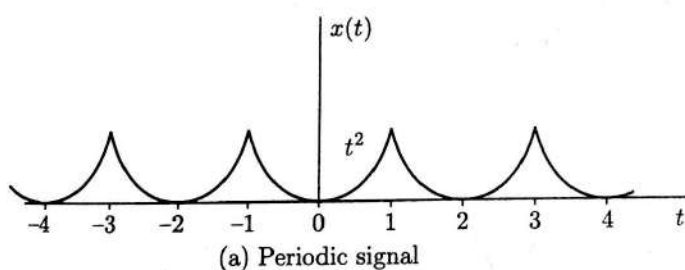


Fig. 3.26

3.4 Find the trigonometric Fourier series for periodic signals shown in Figs 3.27(a) and (b).

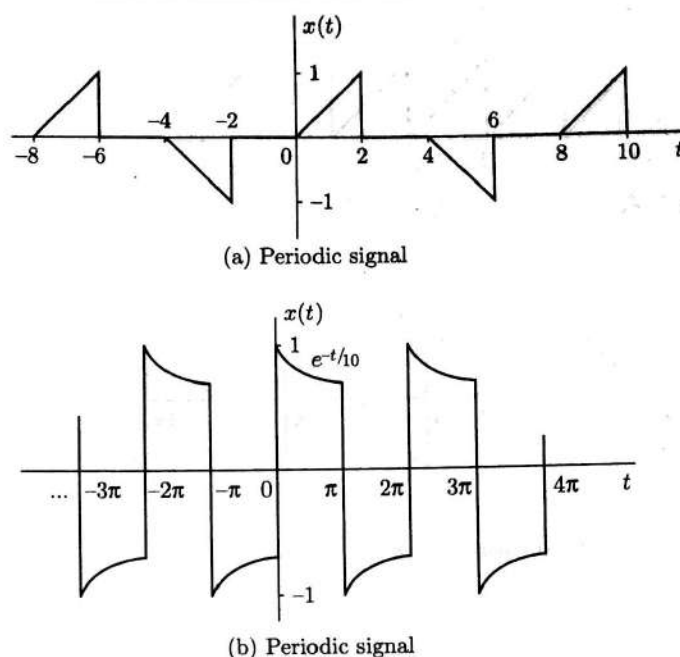


Fig. 3.27

3.5 Consider a continuous-time LTI system with impulse response

$$h(t) = e^{-4|t|}$$

Find the Fourier series representation of the output $y(t)$ for each of the following inputs:

$$(a) x(t) = \sum_{n=-\infty}^{\infty} \delta(t - n)$$

$$(b) x(t) = \sum_{n=-\infty}^{\infty} (-1)^n \delta(t - n)$$

3.6 Let

$$x(t) = \begin{cases} t, & 0 \leq t \leq 1 \\ 2 - t, & 1 \leq t \leq 2 \end{cases}$$

be a periodic signal with fundamental period $T = 2$ and Fourier series coefficients X_n .

(a) Determine the value of X_0 .

(b) Determine the Fourier series representation of $dx(t)/dt$.

(c) Use the result of part (b) and the differentiation property of the Fourier series to determine the Fourier series coefficients of $x(t)$.

3.7 Find the trigonometric Fourier series for the sawtooth wave shown in Fig. 3.28 and plot the line spectrum.

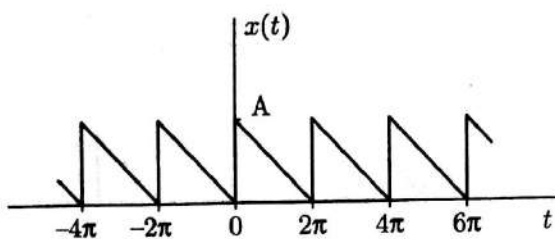


Fig. 3.28 Sawtooth wave

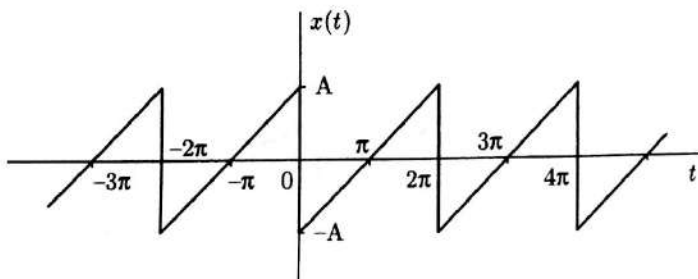


Fig. 3.29 Sawtooth wave

3.8 Find the trigonometric Fourier series for the sawtooth wave shown in Fig. 3.29 and plot the line spectrum.

3.9 Find the trigonometric Fourier series for the waveform shown in Fig. 3.30 and plot the line spectrum.

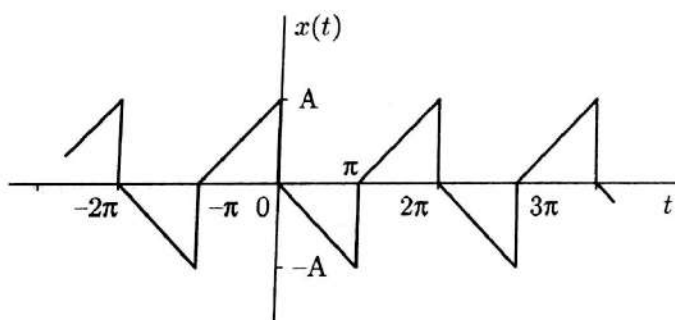


Fig. 3.30 Periodic signal

3.10 Find the trigonometric Fourier series for the waveforms shown in Fig. 3.31.

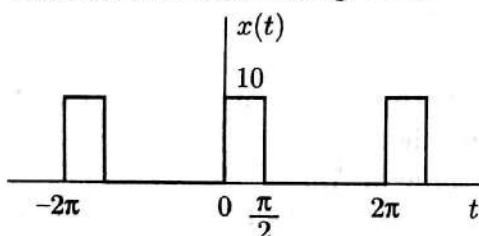


Fig. 3.31 Periodic signal

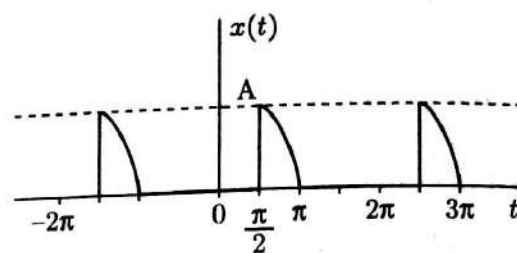


Fig. 3.32 Periodic signal

3.11 Find the trigonometric Fourier series for the waveform shown in Fig. 3.32.

3.12 Find the exponential Fourier series for the waveform shown in Fig. 3.33.

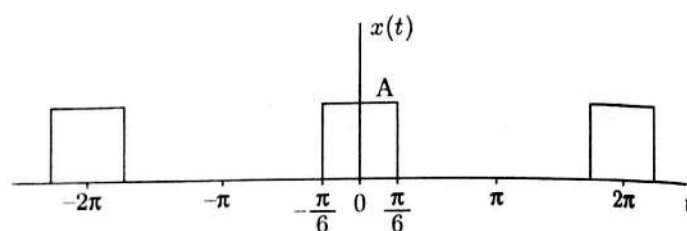


Fig. 3.33 Periodic signal

3.13 The waveform shown in Fig. 3.34 is a trapezoidal in form and of period $T = 4a$. For this periodic function, determine the exponential Fourier series and plot the corresponding magnitude and phase spectra.

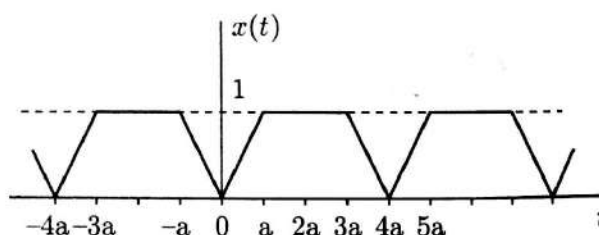


Fig. 3.34 Trapezoidal waveform

3.14 Given the periodic waveform

$$x(t) = t^2, \quad 0 < t < 1$$

Determine the exponential Fourier series and plot the magnitude and phase spectra.

ANSWERS TO MULTIPLE-CHOICE QUESTIONS

1. (b) 2. (d) 3. (a) 4. (a) 5. (c) 6. (a) 7. (b) 8. (a) 9. (d)
10. (b)